

SIMATS ENGINEERING

CSA3703 – SOFTWARE TESTING

ASSIGNMENT 5

Analyze, Compare and Justify Test Measurement, Monitoring, Configuration Management and Review Practices for a Motor Insurance Claims Platform Release

Submission detail	Information
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Register Number	192321040
Department / Section	Information Technology
Course Code and Name	CSA3703 – Software Testing
Assignment Nature	Individual
Duration	1 week
Total Marks	50
Date of Submission	4-11-26
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EXECUTIVE SUMMARY

SureClaim v2.0 release quality-control report

Harbour General Insurance is preparing SureClaim v2.0 with mobile claim intake, photo upload, automated damage estimation, fraud-flag rules, claim settlement payout and a garage-partner portal. The release contains about 22 KLOC of new or changed code, 96 requirements/user stories and 1,400 planned test cases. The supplied tracking data shows that test execution was behind plan from the beginning, although the gap reduced by week 9. Defect discovery was high in the middle of the release and the week-9 position still contains 20 open defects, including three critical defects. Requirement coverage is 90%, statement coverage 78% and branch coverage 64%.

The report compares all three candidate techniques in each task and selects a practical combination. The recommended baseline is a balanced quality-control configuration: a small measurement set, a concise RAG management report, a live operational dashboard, cause-focused schedule control, layered stopping criteria, Git plus configuration baselines and deployment audit, risk-driven reviews, tracked review actions, and a release evidence package. For the changed fortnightly continuous-delivery scenario, this evolves into quality-as-code with signed builds and automatic evidence capture.

Decision area	Preferred practice	Reason
Measurement	Balanced metric set	Shows schedule, quality, coverage and risk.
Status	RAG + live dashboard	Different audiences need different views.
Control	Remove causes first	Fixes bottlenecks instead of hiding them.
Stopping	Layered risk/defect/cost	Prevents false completion.
SCM	Git + baseline + audit	Prevents wrong configuration release.
Reviews	Risk-driven; formal for critical logic	Fits the review-effort limit.
Adaptation	Quality-as-code	Supports fortnightly evidence.

The governing principle is simple: a metric without a decision rule is only a number, a test result without configuration identity is weak evidence, and a review without tracked closure is only a conversation. The report therefore joins measurement, reporting, monitoring, configuration and review into one release decision chain.

REPORT MAP

Evidence-oriented structure

Section	Pages	Evidence

Executive summary and baseline	2–4	Scenario, data and assumptions
Task 1	5–11	Metrics, milestones, GQM, final set
Task 2	12–18	RAG, charts, reporting comparison
Task 3	19–25	Gap analysis and control simulation
Task 4	26–32	Stopping rules, trend and cost
Task 5	33–39	SCM approaches and incident replay
Task 6	40–46	Review types and AI reviewer
Task 7	47–53	Review plans and effort limit
Task 8	54–60	Review reports and follow-up
Task 9	61–68	Integrated configurations and prototype
Task 10	69–76	Fortnightly adaptation
Appendices	77–80	Data, formulas, checklist, sign-off

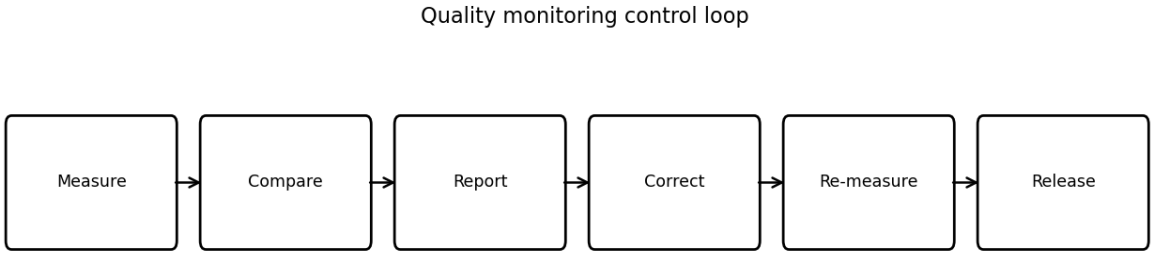


Figure. Measure → report → monitor → control → review → configuration audit → release decision

The report is organised so that each task can be assessed independently while still contributing to the final process. Tables use consistent definitions. Charts use the supplied cumulative tracking data. Simulated execution, review and recovery results are labelled as illustrative because the assignment statement does not contain actual student-run tool output.

ASELINE DATA, ASSUMPTIONS AND EVIDENCE RULES

SureClaim operating baseline

Parameter	Given	Interpretation
Release	12 weeks	Testing W1–W9; readiness W10–W12
Code	22 KLOC	Density denominator
Requirements	96	86 covered by W9
Tests	1,400	Cumulative planned baseline
Team	15	RM, 5 testers, 6 developers, build/config, BA, QA analyst
W9 defects	20 open; 3 critical	Critical defects block recommended release
Coverage	90/78/64%	Requirement/statement/branch
Review limit	10%	540 h under stated project-effort assumption
Defect cost	₹2,500/₹12,000/₹60,000	Review/system/post-release

The project effort assumption is 15 people × 12 weeks × 30 hours per week = 5,400 person-hours. Ten percent gives a maximum review budget of 540 person-hours. Where an hourly monetary rate or productivity uplift is needed, this report labels it as a planning assumption rather than supplied project data. A real submission should replace illustrative values with the student's actual exercise observations when required.

```
# Evidence integrity rule
if status != "Executed":
    result = "NOT RUN"
elif not build_id or not config_hash:
    result = "EVIDENCE INCOMPLETE"
else:
    result = observed_result
```

The same evidence rule applies to configuration and review. A defect is not considered fixed until a retest is linked to the same controlled build. A release is not considered configuration-controlled merely because a filename says final; the exact commit, tag or manifest must be identifiable. AI-generated summaries are drafts until checked against their source records.

TASK 1.1 – Prediction and measurement purpose

This section addresses task 1.1 – prediction and measurement purpose for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction: the schedule gap will be the earliest warning because execution trails plan from week 1, while defect and coverage indicators become more informative as the release matures. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Measure	Definition	W5	W9	Interpretation
Execution %	Executed ÷ planned	75.0%	95.7%	Persistent gap
Pass rate	Passed ÷ executed	81.9%	90.3%	Improves but not sufficient
Schedule variance	Executed − planned	−240	−60	Still behind
Open defects	Found − closed	60	20	Backlog improves
Cumulative density	Found ÷ 22 KLOC	8.09	11.82	W9 exceeds 10/KLOC target

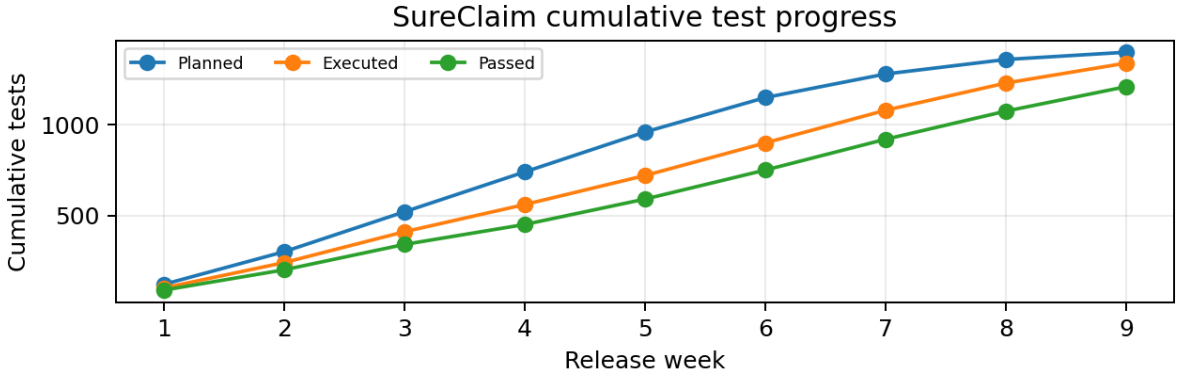


Figure. TASK 1.1 – Prediction and measurement purpose – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 1.2 – Week 5 metric calculation

This section addresses task 1.2 – week 5 metric calculation for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Week 5 is the critical control point because execution is 720/960 = 75%. The pass rate is 590/720 = 81.94%, schedule variance is −240, cumulative defect-find rate is 178/720 = 24.72%, cumulative closure rate is 118/178 = 66.29%, and open backlog is 60. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Measure	Formula	Result	Meaning
Execution	720/960	75.00%	Red under >15% gap rule
Pass	590/720	81.94%	Failures remain
Find rate	178/720	24.72%	High discovery
Closure	118/178	66.29%	Closure lags discovery
Open	178−118	60	High pressure
Density	178/22	8.09/KLOC	Below target if cumulative found is used

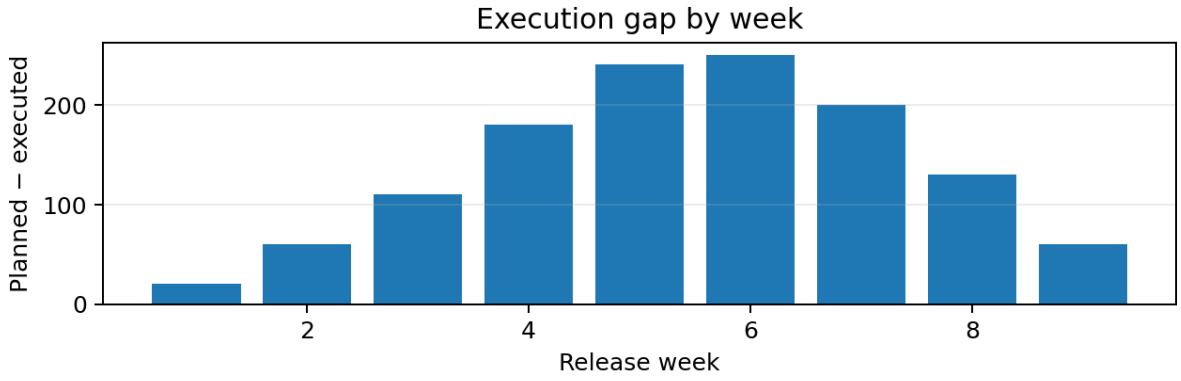


Figure. TASK 1.2 – Week 5 metric calculation – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 1.3 – Week 9 metric calculation

This section addresses task 1.3 – week 9 metric calculation for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Week 9 reaches 1,340/1,400 = 95.71% execution and 1,210/1,340 = 90.30% pass. Schedule variance is −60; cumulative find rate is 19.40%; closure rate is 92.31%; open backlog is 20; cumulative density is 11.82/KLOC. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Indicator	Calculation	Result	Interpretation
Execution	1340/1400	95.71%	Near plan
Pass	1210/1340	90.30%	Below 95% example gate
Open	260−240	20	Includes 3 critical
Density	260/22	11.82/KLOC	Above target
Open density	20/22	0.91/KLOC	Not a replacement for total density

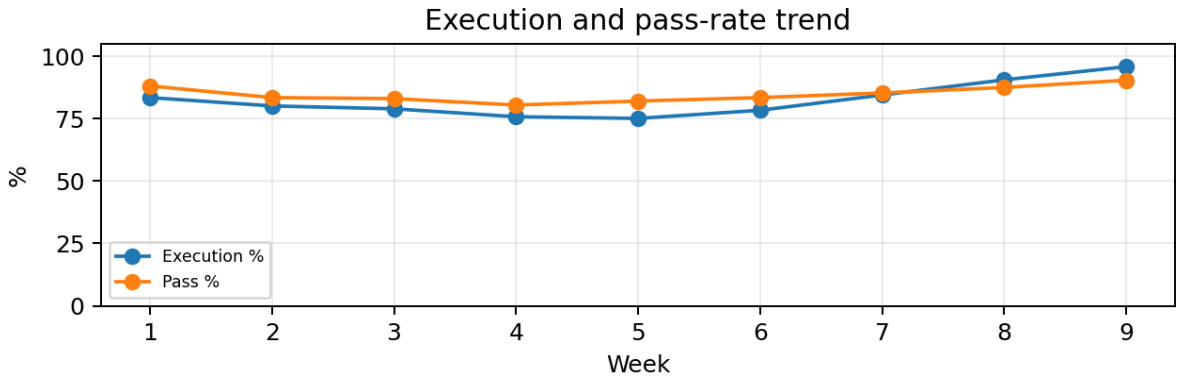


Figure. TASK 1.3 – Week 9 metric calculation – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 1.4 – Milestones and achievement status

This section addresses task 1.4 – milestones and achievement status for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Five measurable milestones convert the data into control points. A milestone is met only when the supplied data supports its condition; vague statements such as testing is going well are not acceptable. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Milestone	Condition	Observed	Status
M1 W2	≥80% execution	80%	Met
M2 W4	≥75% execution	75.7%	Met
M3 W7	≥90% execution	84.4%	Not met
M4 W9	≥95% + zero critical	95.7% + 3 critical	Not met
M5 W10	Candidate configuration audited	Not supplied	Evidence required

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 1.5 – First warning and leading indicators

This section addresses task 1.5 – first warning and leading indicators for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

The first schedule signal appears in week 1: 100 of 120 tests were executed, a 16.67% shortfall. Under the example RAG rule of red above 15% shortfall, the warning is immediate. The widening gap is therefore a trend rather than a week-9 surprise. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Week	Gap	Gap %	Signal
1	20	16.7%	Red
2	60	20.0%	Red
3	110	21.2%	Red
4	180	24.3%	Red
5	240	25.0%	Red
6	250	21.7%	Red
7	200	15.6%	Red
8	130	9.6%	Amber
9	60	4.3%	Green for schedule only

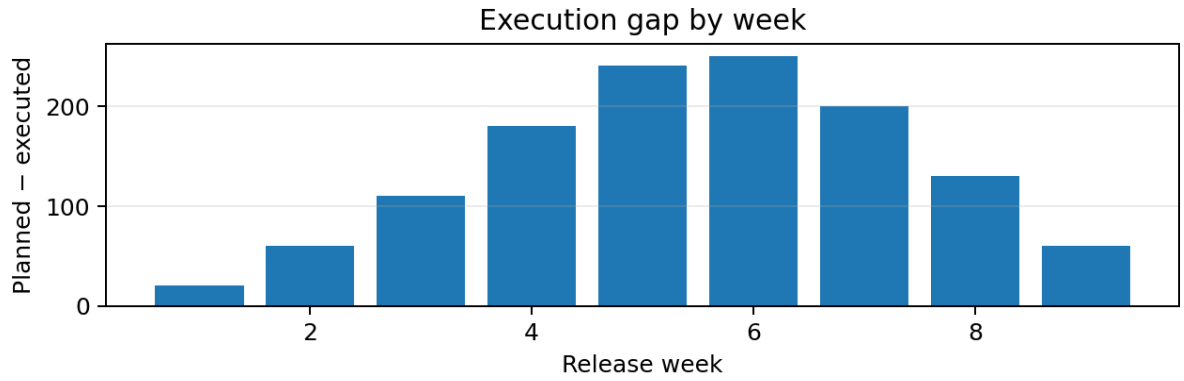


Figure. TASK 1.5 – First warning and leading indicators – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 1.6 – Misuse of measurement approaches

This section addresses task 1.6 – misuse of measurement approaches for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Plan measures can be gamed by changing denominators; defect counts can reward quantity rather than value; coverage can create false confidence when low-risk code is covered while high-risk requirements remain weak. Metrics must therefore be balanced and linked to risk. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Approach	Misuse	Example	Control
Plan/actual	Denominator manipulation	Defer hard tests	Freeze baseline
Defect	Count gaming	Duplicates	Use severity/escape rate
Coverage	Coverage without assertions	Low-risk code	Risk-weight coverage
All	Metric becomes target	Optimise number	Use balanced scorecard

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 1.7 – GQM and final measurement set

This section addresses task 1.7 – gqm and final measurement set for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

The goal is release SureClaim with no open critical defect. Questions address critical risk, test progress, defect flow, requirement coverage and configuration identity. The final set contains no more than eight metrics. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Level	Example
Goal	Release without open critical defect
Question	Are critical defects open/aging?
Metric	Open critical defects and age
Question	Is scope progressing?
Metric	Execution % and schedule variance
Question	Is quality improving?
Metric	Defect arrival/closure
Question	Is risk covered?
Metric	Requirement/branch coverage
Question	Is candidate trustworthy?
Metric	Build/config identity

Goal–Question–Metric chain



Figure. TASK 1.7 – GQM and final measurement set – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 2.1 – Prediction and comparison of reporting formats

This section addresses task 2.1 – prediction and comparison of reporting formats for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use RAG for management plus a live dashboard for the test team.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Format	Strength	Weakness	Effort
RAG	Fast decision	Hides detail	Low
Chart pack	Trend context	More reading	Medium
Live dashboard	Current detail	Needs narrative	Medium

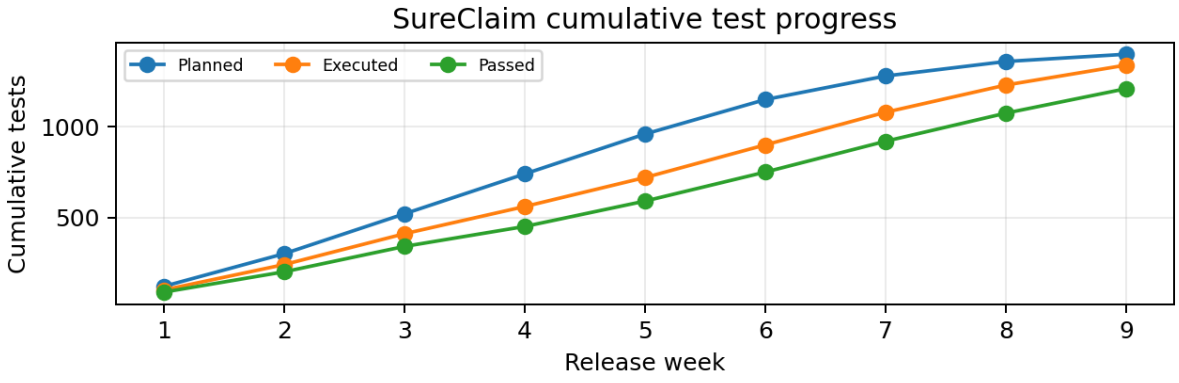


Figure. TASK 2.1 – Prediction and comparison of reporting formats – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 2.2 – Week 6 RAG report

This section addresses task 2.2 – week 6 rag report for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

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Area	Week 6	RAG	Reason
Schedule	900/1150=78.3%	Red	21.7% gap
Quality	214 found/156 closed	Amber	Backlog high
Coverage	Not supplied W6	Amber	Evidence gap
Configuration	Not supplied	Red	Identity must be verified

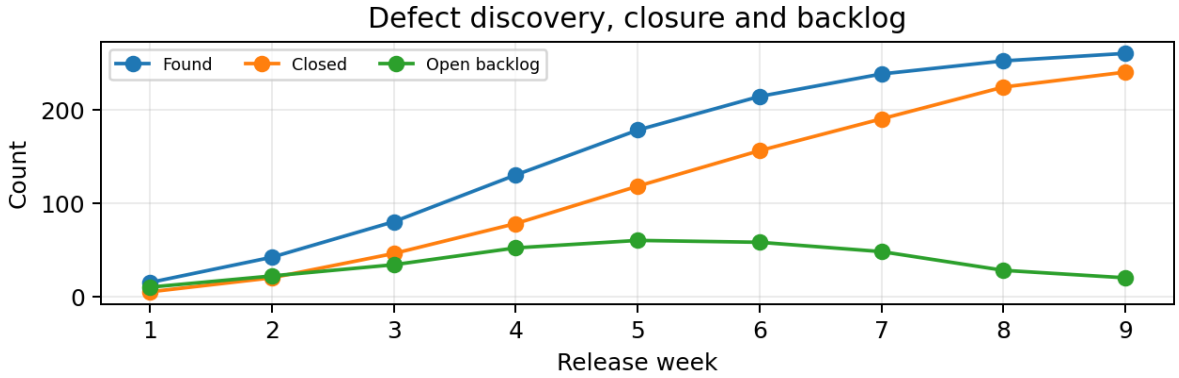


Figure. TASK 2.2 – Week 6 RAG report – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 2.3 – Metric-chart pack design

This section addresses task 2.3 – metric-chart pack design for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

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Chart	Question	Signal
Progress	Work complete?	900 vs 1150
Defects	Quality flow?	214 vs 156
Gap	Schedule risk?	250 tests
Pass	Executed quality?	83.3%

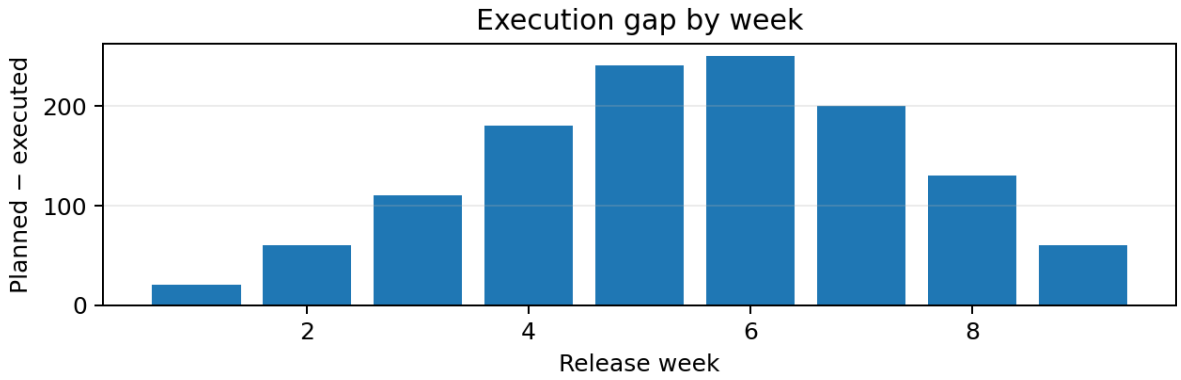


Figure. TASK 2.3 – Metric-chart pack design – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 2.4 – RAG rules for weeks 3, 6 and 9

This section addresses task 2.4 – rag rules for weeks 3, 6 and 9 for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

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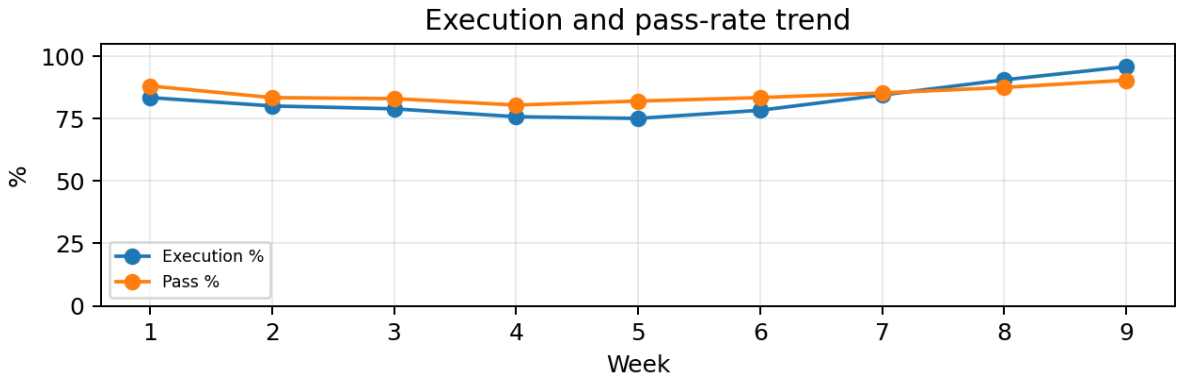


Figure. TASK 2.4 – RAG rules for weeks 3, 6 and 9 – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 2.5 – Audience, effort and false comfort

This section addresses task 2.5 – audience, effort and false comfort for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

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Area	Week 6	RAG	Reason
Schedule	900/1150=78.3%	Red	21.7% gap
Quality	214 found/156 closed	Amber	Backlog high
Coverage	Not supplied W6	Amber	Evidence gap
Configuration	Not supplied	Red	Identity must be verified

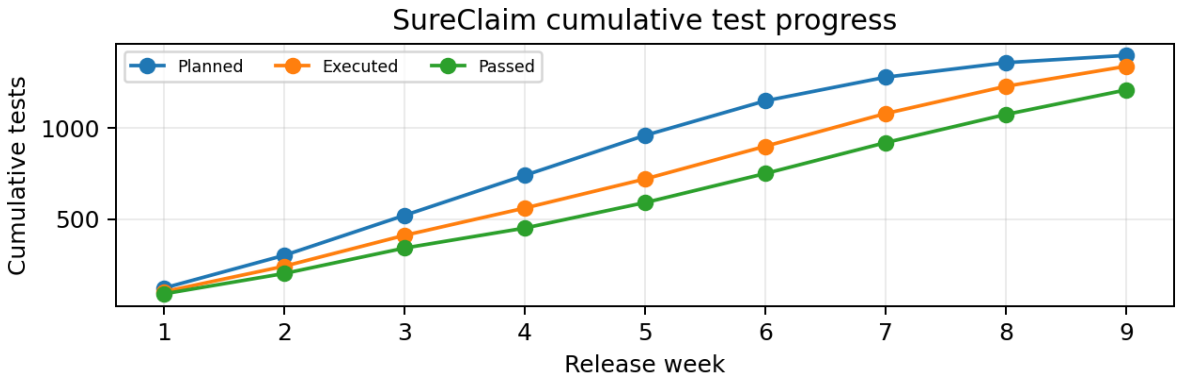


Figure. TASK 2.5 – Audience, effort and false comfort – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 2.6 – LLM-generated report critique

This section addresses task 2.6 – llm-generated report critique for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

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Gap	Schedule risk?	250 tests
Pass	Executed quality?	83.3%

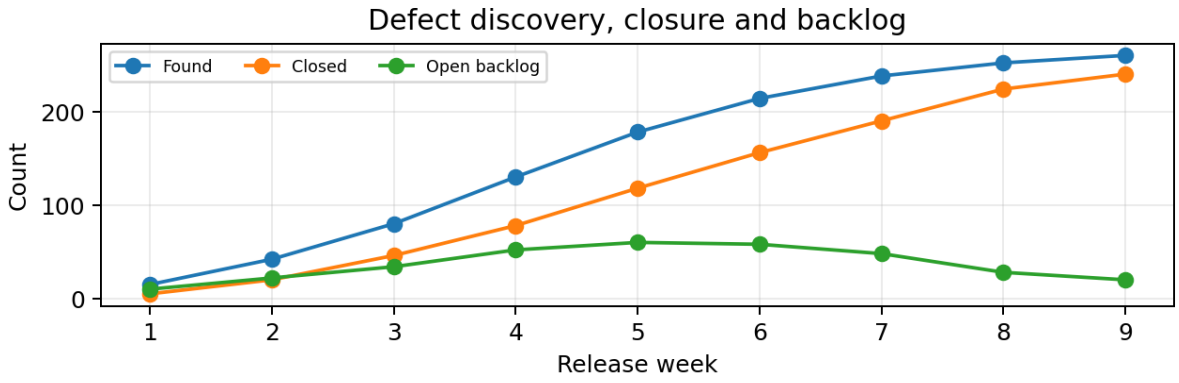


Figure. TASK 2.6 – LLM-generated report critique – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 2.7 – Final reporting selection

This section addresses task 2.7 – final reporting selection for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use RAG for management plus a live dashboard for the test team.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Format	Strength	Weakness	Effort
RAG	Fast decision	Hides detail	Low
Chart pack	Trend context	More reading	Medium
Live dashboard	Current detail	Needs narrative	Medium

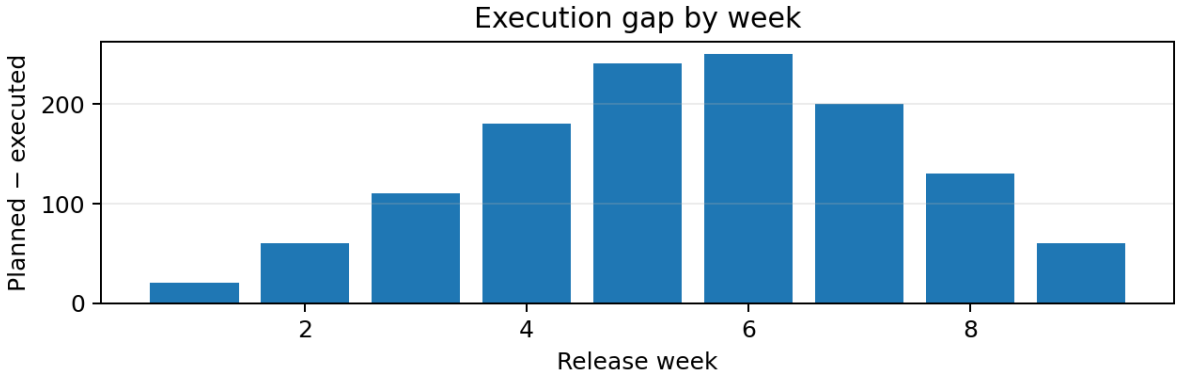


Figure. TASK 2.7 – Final reporting selection – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 3.1 – Prediction and comparison of control responses

This section addresses task 3.1 – prediction and comparison of control responses for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Remove the cause first, then use targeted extra capacity; descope only low-risk work.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Response	Benefit	Risk	Control
Add effort	Capacity	Onboarding	Use after bottleneck
Descope	Schedule	Coverage loss	Risk rank
Remove cause	Sustainable	Diagnosis needed	Root-cause action

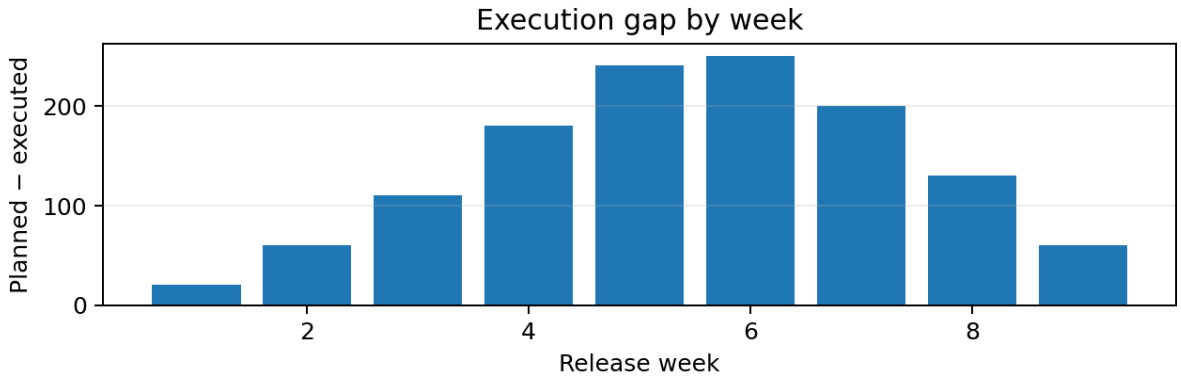


Figure. TASK 3.1 – Prediction and comparison of control responses – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 3.2 – Weekly execution gap calculation

This section addresses task 3.2 – weekly execution gap calculation for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Remove the cause first, then use targeted extra capacity; descope only low-risk work.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Week	Planned	Executed	Gap
1	120	100	20
2	300	240	60
3	520	410	110
4	740	560	180
5	960	720	240
6	1150	900	250
7	1280	1080	200
8	1360	1230	130
9	1400	1340	60

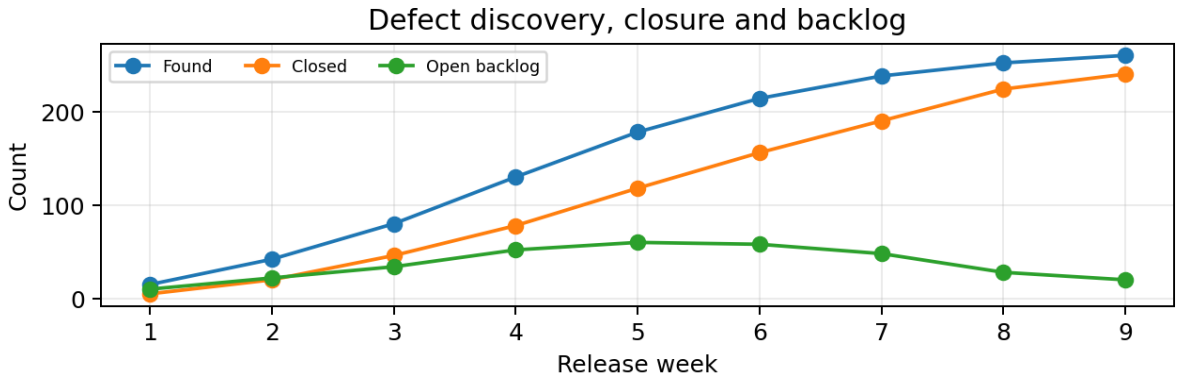


Figure. TASK 3.2 – Weekly execution gap calculation – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 3.3 – Projection assumptions for the three responses

This section addresses task 3.3 – projection assumptions for the three responses for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Remove the cause first, then use targeted extra capacity; descope only low-risk work.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Simulation	Assumption	Expected effect
A	+60% after training	Higher throughput
B	Drop low-risk 20%	Lower scope
C	+25% then +35%	Root-cause recovery

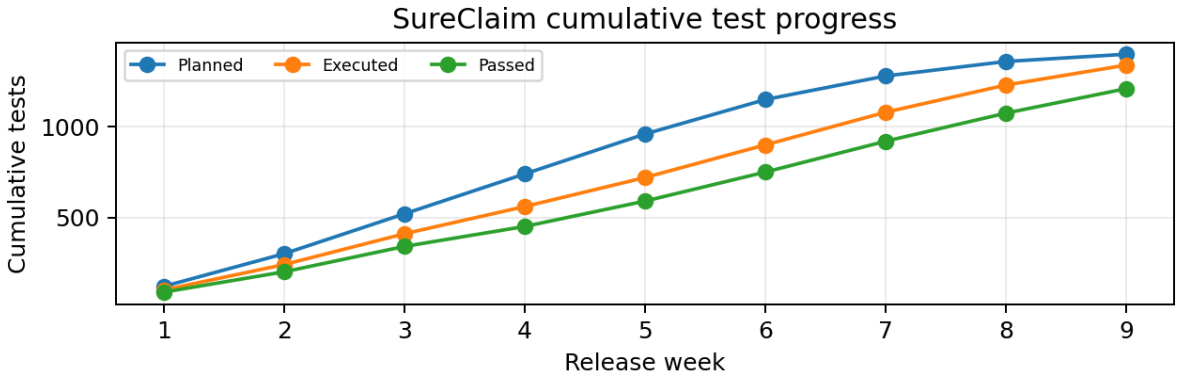


Figure. TASK 3.3 – Projection assumptions for the three responses – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 3.4 – Add-effort simulation

This section addresses task 3.4 – add-effort simulation for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Remove the cause first, then use targeted extra capacity; descope only low-risk work.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Response	Benefit	Risk	Control
Add effort	Capacity	Onboarding	Use after bottleneck
Descope	Schedule	Coverage loss	Risk rank
Remove cause	Sustainable	Diagnosis needed	Root-cause action

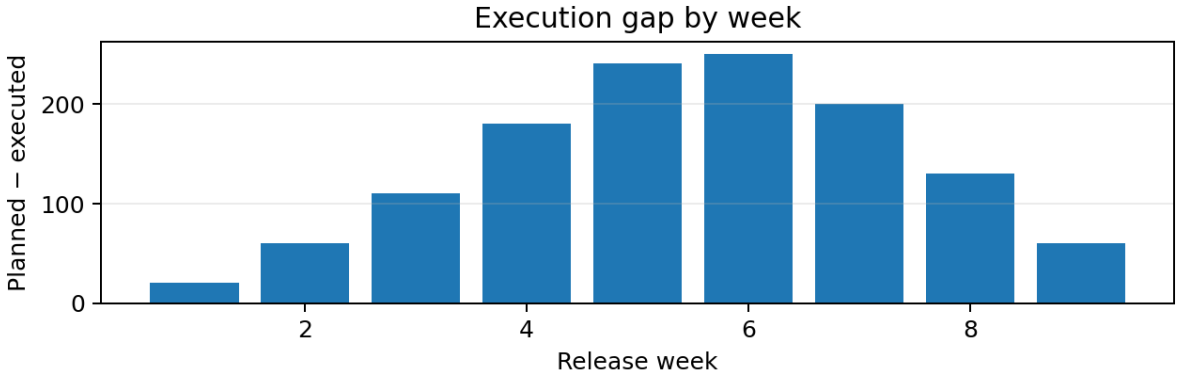


Figure. TASK 3.4 – Add-effort simulation – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 3.5 – Re-prioritisation and residual risk

This section addresses task 3.5 – re-prioritisation and residual risk for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Remove the cause first, then use targeted extra capacity; descope only low-risk work.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Week	Planned	Executed	Gap
1	120	100	20
2	300	240	60
3	520	410	110
4	740	560	180
5	960	720	240
6	1150	900	250
7	1280	1080	200
8	1360	1230	130
9	1400	1340	60

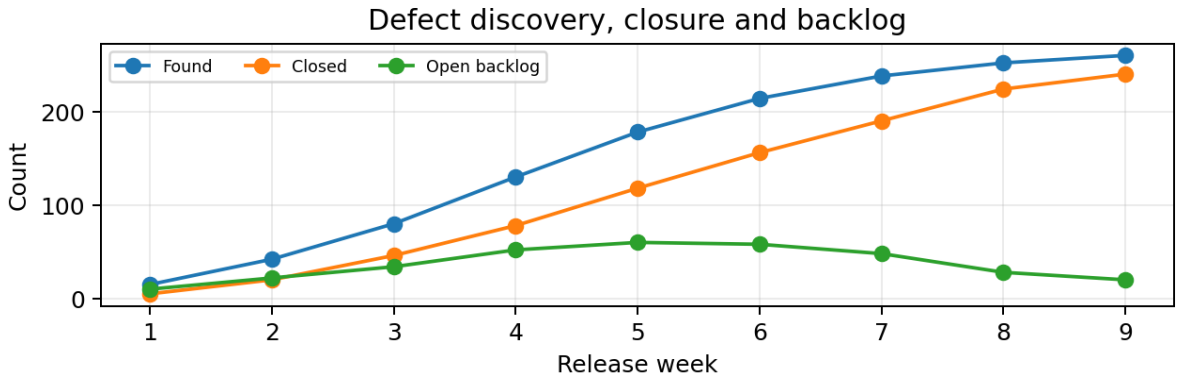


Figure. TASK 3.5 – Re-prioritisation and residual risk – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 3.6 – Remove-cause response and backlog interpretation

This section addresses task 3.6 – remove-cause response and backlog interpretation for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Remove the cause first, then use targeted extra capacity; descope only low-risk work.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Simulation	Assumption	Expected effect
A	+60% after training	Higher throughput
B	Drop low-risk 20%	Lower scope
C	+25% then +35%	Root-cause recovery

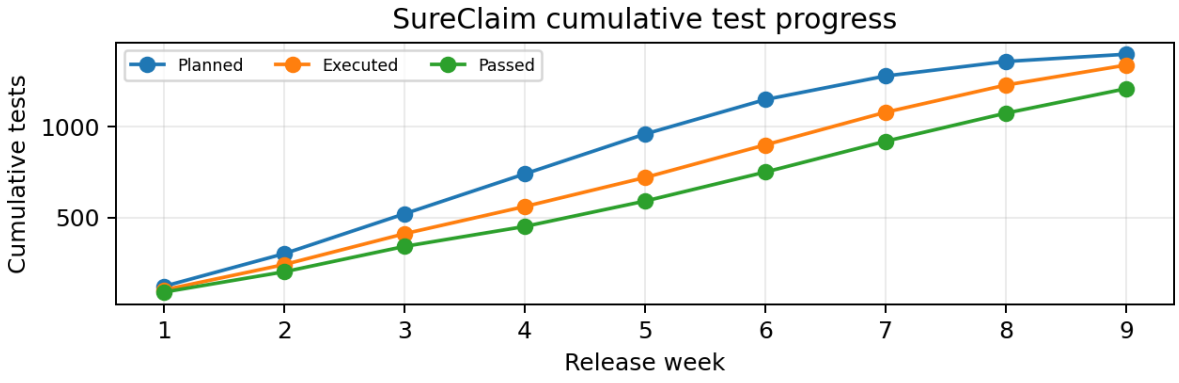


Figure. TASK 3.6 – Remove-cause response and backlog interpretation – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 3.7 – Final control decision

This section addresses task 3.7 – final control decision for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Remove the cause first, then use targeted extra capacity; descope only low-risk work.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Response	Benefit	Risk	Control
Add effort	Capacity	Onboarding	Use after bottleneck
Descope	Schedule	Coverage loss	Risk rank
Remove cause	Sustainable	Diagnosis needed	Root-cause action

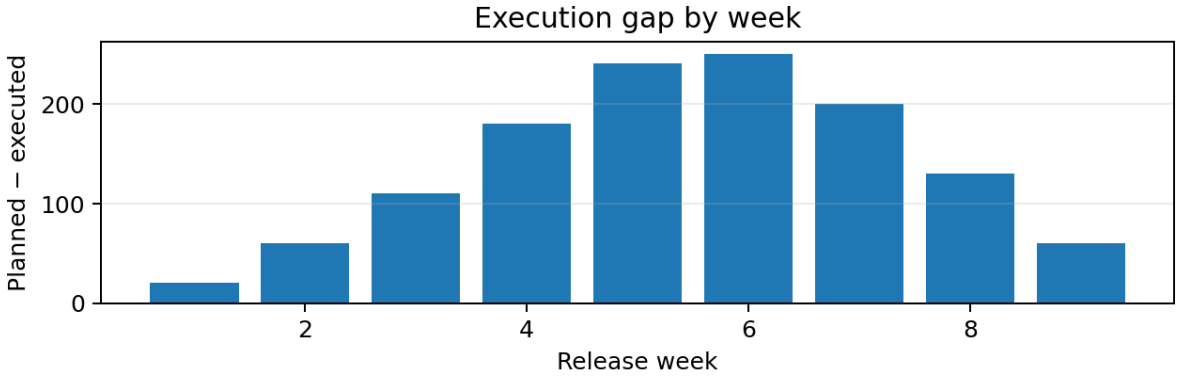


Figure. TASK 3.7 – Final control decision – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 4.1 – Prediction and stopping philosophy

This section addresses task 4.1 – prediction and stopping philosophy for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use layered safety gates plus defect trend, coverage and residual-cost evidence.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Criterion	Rule	W9	Decision
A	100% + 95% pass	95.7/90.3	Not met
B	<10 defects 2 weeks + 0 critical	8 defects but 3 critical	Not met
C	Expected loss < testing cost	Continue model	Continue

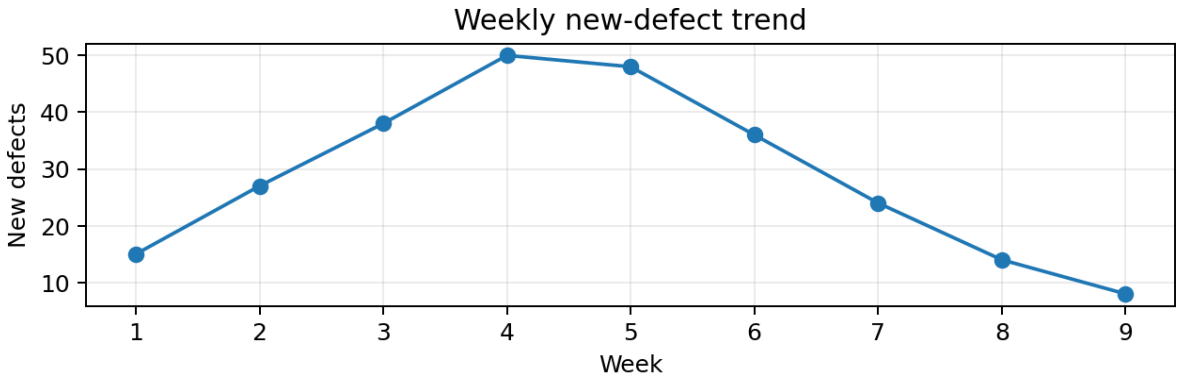


Figure. TASK 4.1 – Prediction and stopping philosophy – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 4.2 – Criterion A: test-case completion

This section addresses task 4.2 – criterion a: test-case completion for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use layered safety gates plus defect trend, coverage and residual-cost evidence.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Scenario	Remaining defects	Use
Optimistic	4	Sensitivity
Planning	8	Main case
Pessimistic	12	Stress case

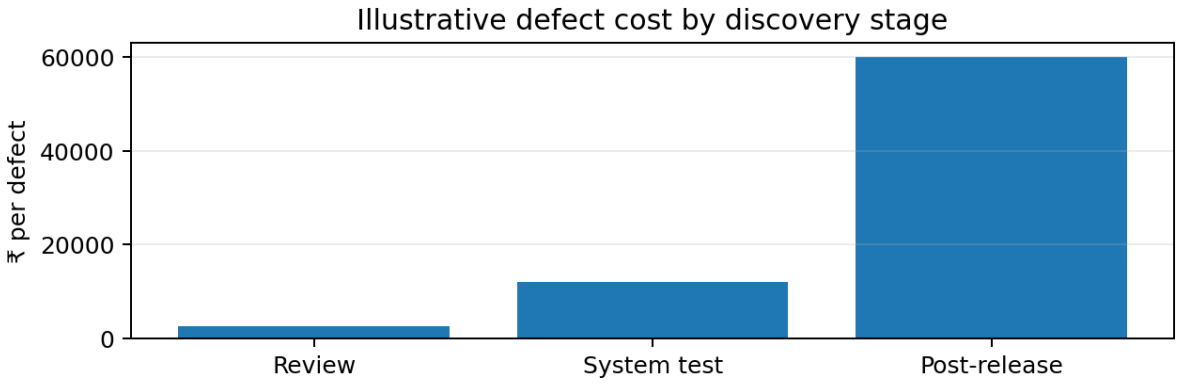


Figure. TASK 4.2 – Criterion A: test-case completion – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 4.3 – Criterion B: defect-trend stopping

This section addresses task 4.3 – criterion b: defect-trend stopping for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use layered safety gates plus defect trend, coverage and residual-cost evidence.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Gate	Threshold	Evidence
Critical	0 open	Defect tool
Requirements	≥95%; critical 100%	Traceability
Execution	≥98%; critical 100%	Test tool
Branch	≥70%; higher critical	Coverage
Trend	<10 for 2 weeks	Chart

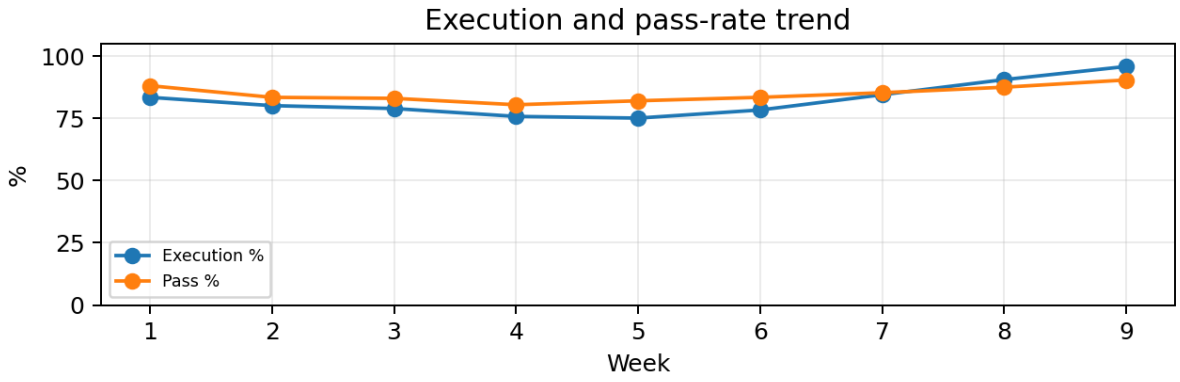


Figure. TASK 4.3 – Criterion B: defect-trend stopping – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 4.4 – Residual defect estimate

This section addresses task 4.4 – residual defect estimate for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use layered safety gates plus defect trend, coverage and residual-cost evidence.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Criterion	Rule	W9	Decision
A	100% + 95% pass	95.7/90.3	Not met
B	<10 defects 2 weeks + 0 critical	8 defects but 3 critical	Not met
C	Expected loss < testing cost	Continue model	Continue

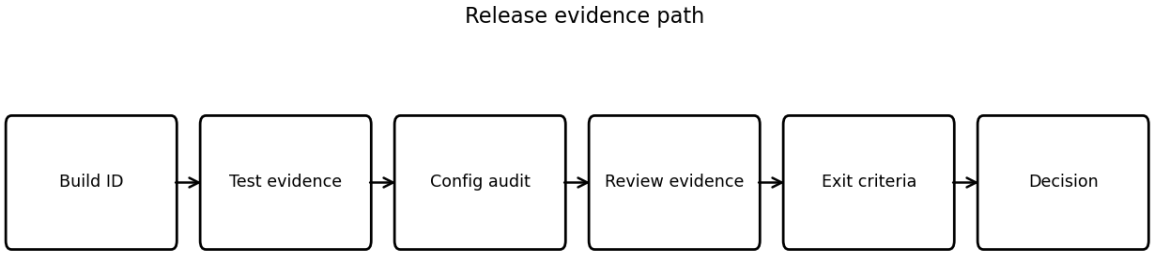


Figure. TASK 4.4 – Residual defect estimate – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 4.5 – Risk-and-cost stopping model

This section addresses task 4.5 – risk-and-cost stopping model for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use layered safety gates plus defect trend, coverage and residual-cost evidence.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Scenario	Remaining defects	Use
Optimistic	4	Sensitivity
Planning	8	Main case
Pessimistic	12	Stress case

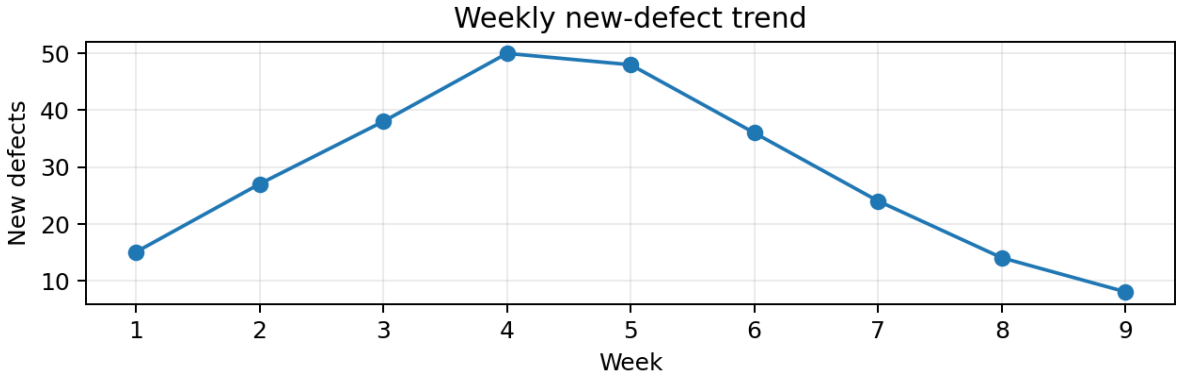


Figure. TASK 4.5 – Risk-and-cost stopping model – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 4.6 – Weaknesses of the three criteria

This section addresses task 4.6 – weaknesses of the three criteria for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use layered safety gates plus defect trend, coverage and residual-cost evidence.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Gate	Threshold	Evidence
Critical	0 open	Defect tool
Requirements	≥95%; critical 100%	Traceability
Execution	≥98%; critical 100%	Test tool
Branch	≥70%; higher critical	Coverage
Trend	<10 for 2 weeks	Chart

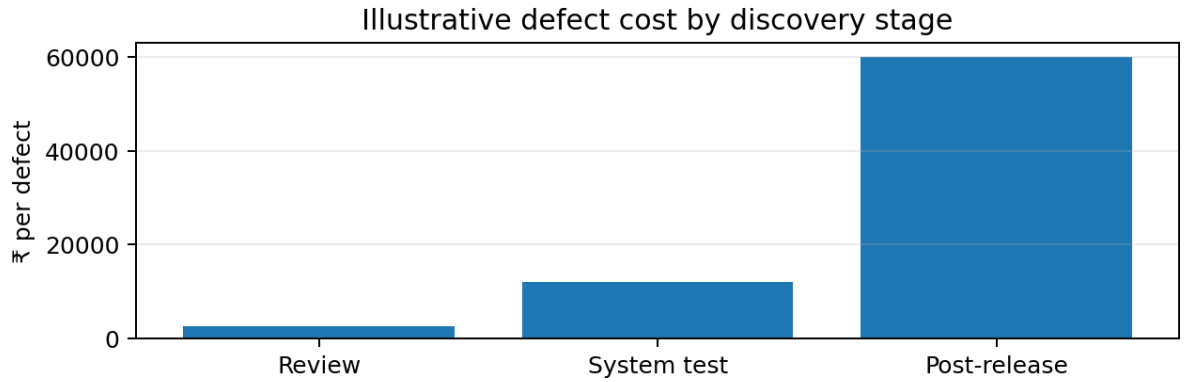


Figure. TASK 4.6 – Weaknesses of the three criteria – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 4.7 – Final layered stopping policy

This section addresses task 4.7 – final layered stopping policy for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use layered safety gates plus defect trend, coverage and residual-cost evidence.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Criterion	Rule	W9	Decision
A	100% + 95% pass	95.7/90.3	Not met
B	<10 defects 2 weeks + 0 critical	8 defects but 3 critical	Not met
C	Expected loss < testing cost	Continue model	Continue

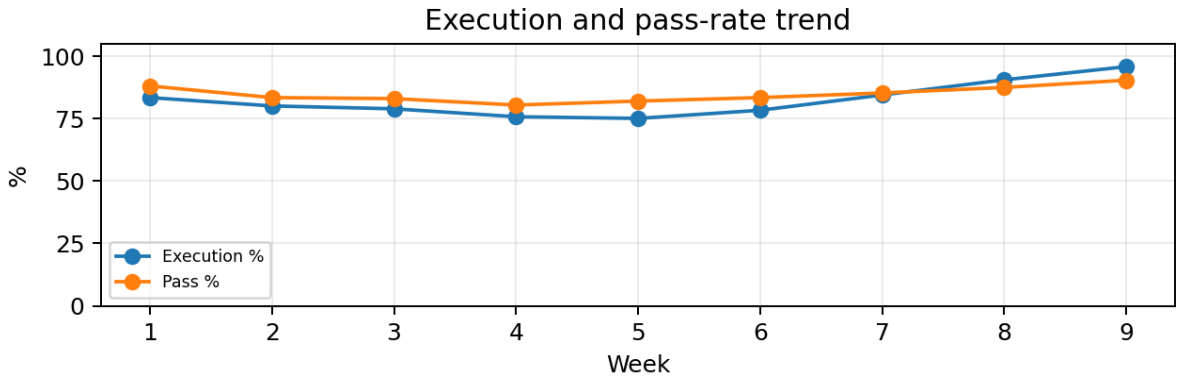


Figure. TASK 4.7 – Final layered stopping policy – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 5.1 – Prediction and comparison of SCM approaches

This section addresses task 5.1 – prediction and comparison of scm approaches for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Adopt Git plus formal baselines and automated deployment comparison for controlled items.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Approach	Traceability	Reproduction	Effort
Shared folder	Low	Weak	Low
Git	High	High	Medium
Full SCM	Very high	Very high	Higher

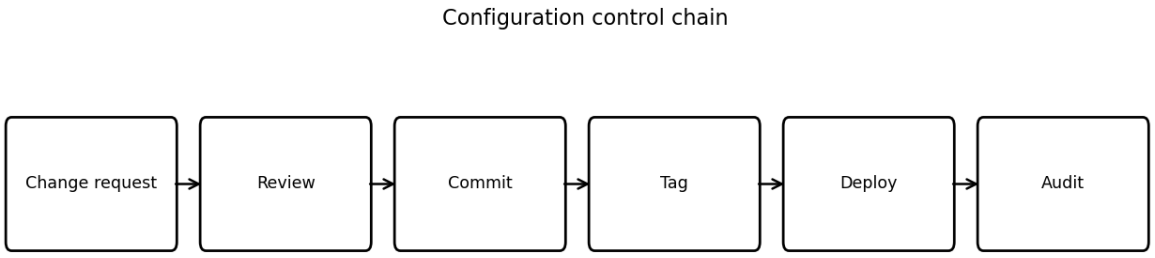


Figure. TASK 5.1 – Prediction and comparison of SCM approaches – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 5.2 – Shared-folder versioning

This section addresses task 5.2 – shared-folder versioning for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Adopt Git plus formal baselines and automated deployment comparison for controlled items.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Item	Control	Evidence
Source	Protected Git	Commit SHA
fraud_rules.json	Baseline + approval	Hash
Premium table	Baseline	Version/hash
Build scripts	Git review	Commit
Test data	Versioned package	Dataset ID

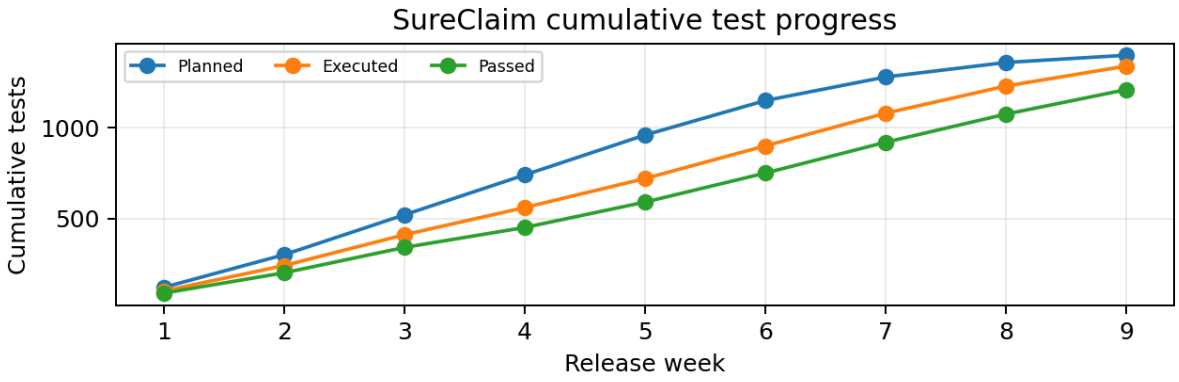


Figure. TASK 5.2 – Shared-folder versioning – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 5.3 – Git-based control

This section addresses task 5.3 – git-based control for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Adopt Git plus formal baselines and automated deployment comparison for controlled items.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Incident	Detection	Recovery
Shared folder	Manual	Find copy
Git	Build comparison	Checkout tag
Full SCM	Automated audit	Restore baseline

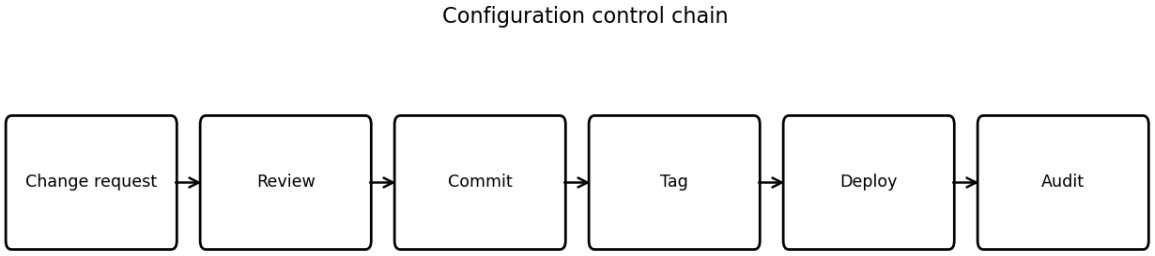


Figure. TASK 5.3 – Git-based control – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 5.4 – Full SCM with baselines and audit

This section addresses task 5.4 – full scm with baselines and audit for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Adopt Git plus formal baselines and automated deployment comparison for controlled items.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Approach	Traceability	Reproduction	Effort
Shared folder	Low	Weak	Low
Git	High	High	Medium
Full SCM	Very high	Very high	Higher

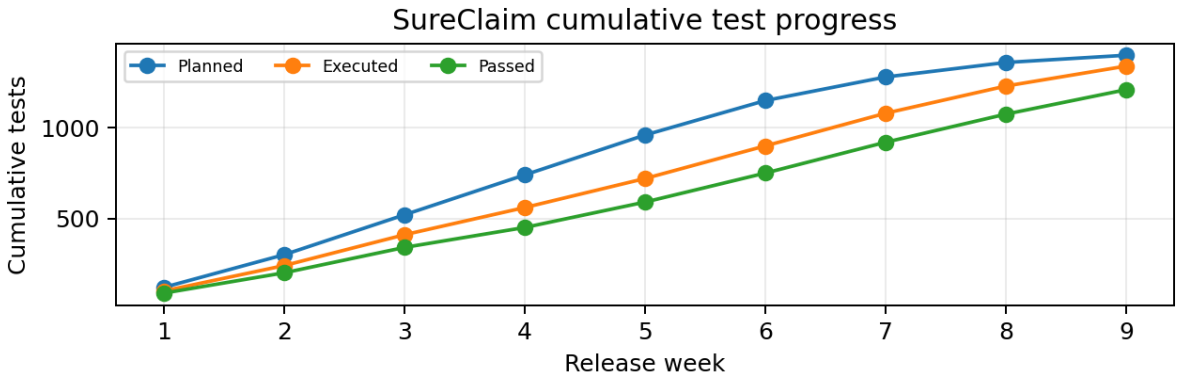


Figure. TASK 5.4 – Full SCM with baselines and audit – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 5.5 – Incident replay and recovery

This section addresses task 5.5 – incident replay and recovery for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Adopt Git plus formal baselines and automated deployment comparison for controlled items.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Item	Control	Evidence
Source	Protected Git	Commit SHA
fraud_rules.json	Baseline + approval	Hash
Premium table	Baseline	Version/hash
Build scripts	Git review	Commit
Test data	Versioned package	Dataset ID

Configuration control chain

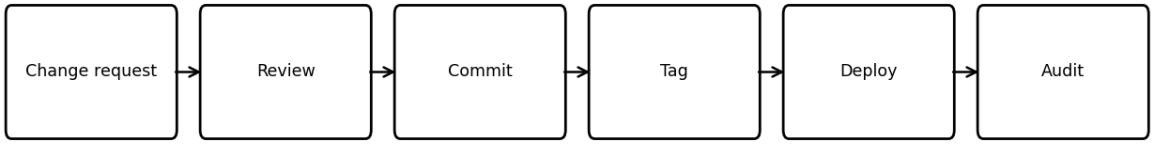


Figure. TASK 5.5 – Incident replay and recovery – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 5.6 – Configuration identity and testing

This section addresses task 5.6 – configuration identity and testing for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Adopt Git plus formal baselines and automated deployment comparison for controlled items.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Incident	Detection	Recovery
Shared folder	Manual	Find copy
Git	Build comparison	Checkout tag
Full SCM	Automated audit	Restore baseline

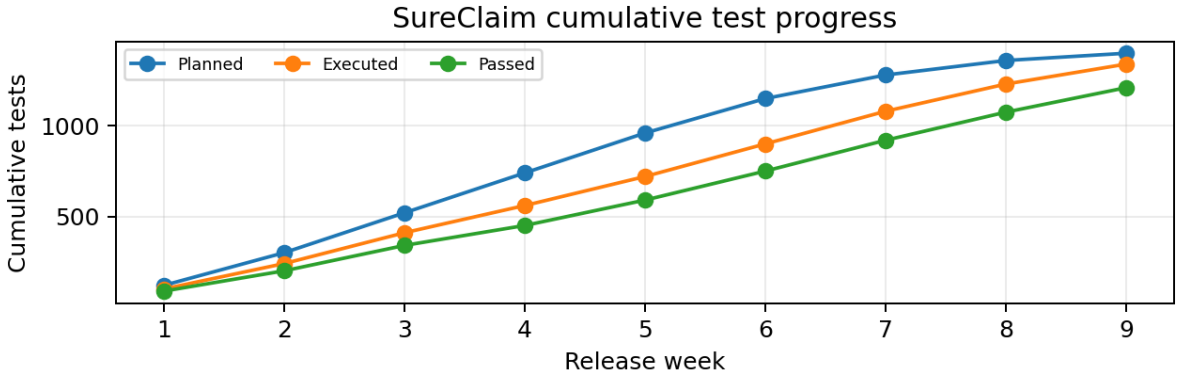


Figure. TASK 5.6 – Configuration identity and testing – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 5.7 – Final SCM choice and AI configuration critique

This section addresses task 5.7 – final scm choice and ai configuration critique for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Adopt Git plus formal baselines and automated deployment comparison for controlled items.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Approach	Traceability	Reproduction	Effort
Shared folder	Low	Weak	Low
Git	High	High	Medium
Full SCM	Very high	Very high	Higher

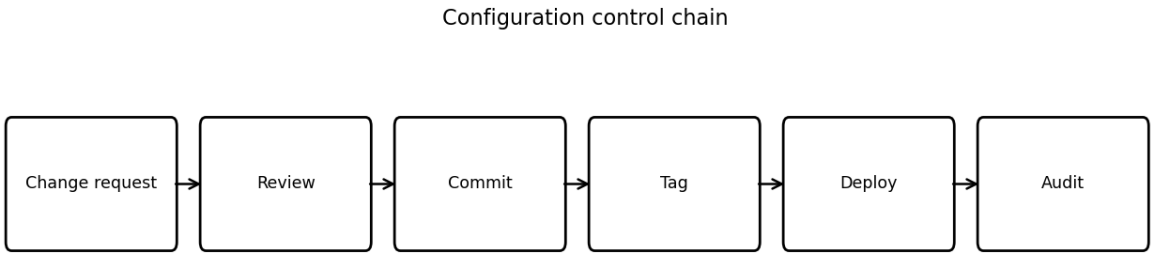


Figure. TASK 5.7 – Final SCM choice and AI configuration critique – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 6.1 – Prediction and comparison of review types

This section addresses task 6.1 – prediction and comparison of review types for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use formal inspection for high-impact fraud/payout/security logic and lighter reviews elsewhere.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Review	Expected detection	Effort	Use
Desk-check	Low/medium	Low	Routine
Technical	Medium/high	Medium	Complex work
Inspection	High	High	Critical logic

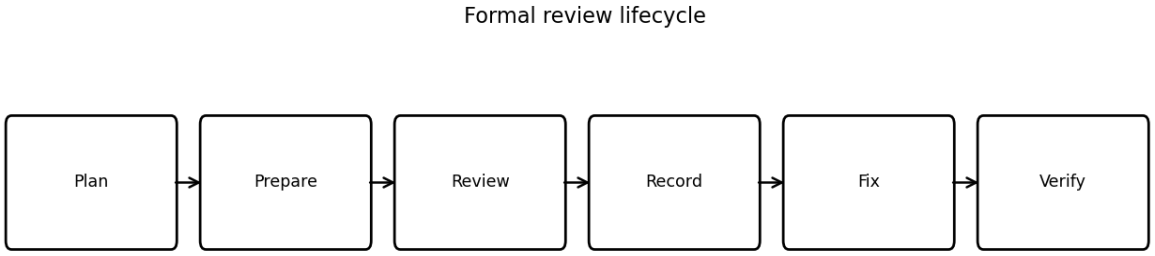


Figure. TASK 6.1 – Prediction and comparison of review types – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 6.2 – Seeded work product and measurement model

This section addresses task 6.2 – seeded work product and measurement model for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use formal inspection for high-impact fraud/payout/security logic and lighter reviews elsewhere.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Measure	Illustrative value	Formula
Seeded defects	8	Controlled exercise
Found	4/6/7	By review type
Detection	50/75/87.5%	Found/8
Time	1/1.5/2 h	Recorded

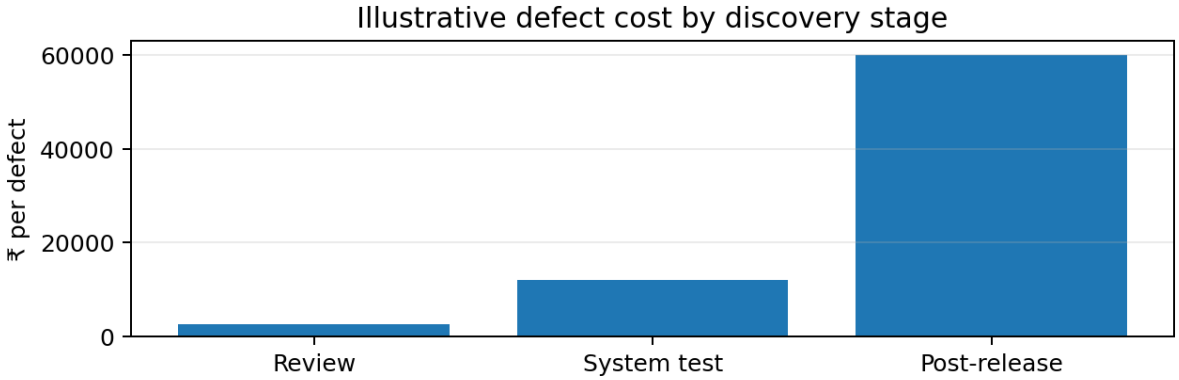


Figure. TASK 6.2 – Seeded work product and measurement model – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 6.3 – Cost saving of early review

This section addresses task 6.3 – cost saving of early review for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use formal inspection for high-impact fraud/payout/security logic and lighter reviews elsewhere.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Cost	₹/defect	Saving vs review
Review	2,500	Baseline
System	12,000	9,500
Post-release	60,000	57,500

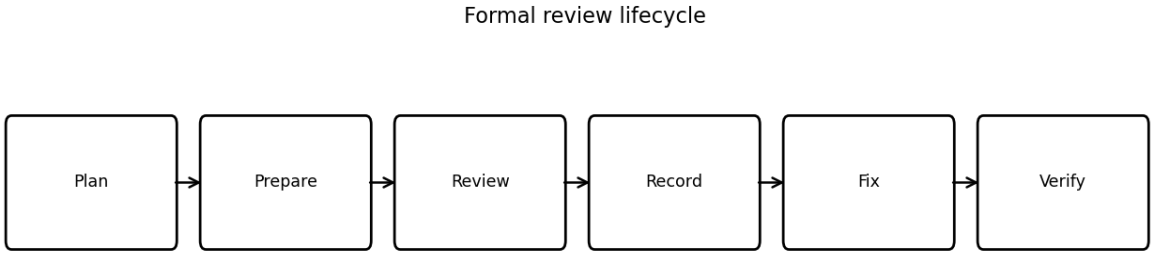


Figure. TASK 6.3 – Cost saving of early review – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 6.4 – Defect types each review detects well

This section addresses task 6.4 – defect types each review detects well for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use formal inspection for high-impact fraud/payout/security logic and lighter reviews elsewhere.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Review	Expected detection	Effort	Use
Desk-check	Low/medium	Low	Routine
Technical	Medium/high	Medium	Complex work
Inspection	High	High	Critical logic

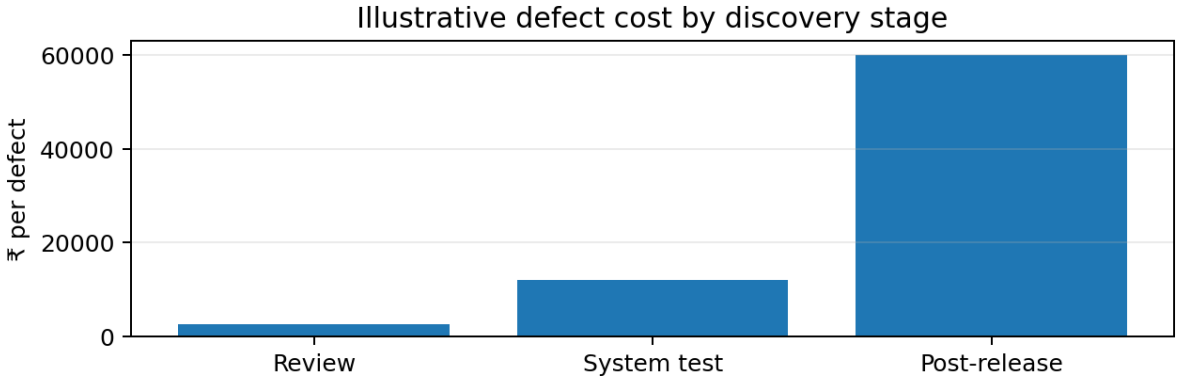


Figure. TASK 6.4 – Defect types each review detects well – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 6.5 – Human and organisational problems

This section addresses task 6.5 – human and organisational problems for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use formal inspection for high-impact fraud/payout/security logic and lighter reviews elsewhere.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Measure	Illustrative value	Formula
Seeded defects	8	Controlled exercise
Found	4/6/7	By review type
Detection	50/75/87.5%	Found/8
Time	1/1.5/2 h	Recorded

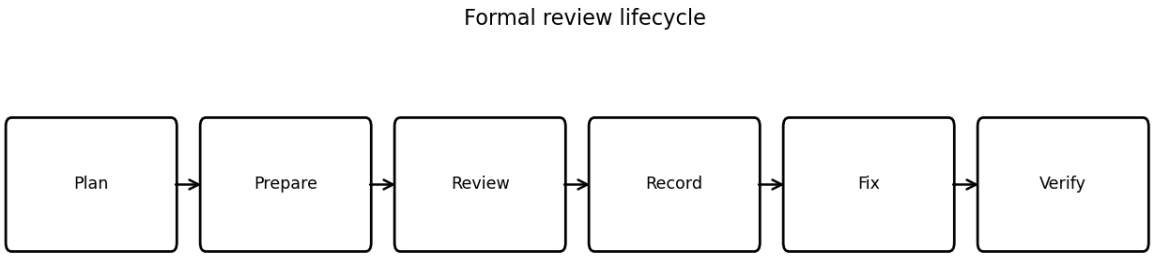


Figure. TASK 6.5 – Human and organisational problems – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 6.6 – AI as additional reviewer

This section addresses task 6.6 – ai as additional reviewer for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use formal inspection for high-impact fraud/payout/security logic and lighter reviews elsewhere.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Cost	₹/defect	Saving vs review
Review	2,500	Baseline
System	12,000	9,500
Post-release	60,000	57,500

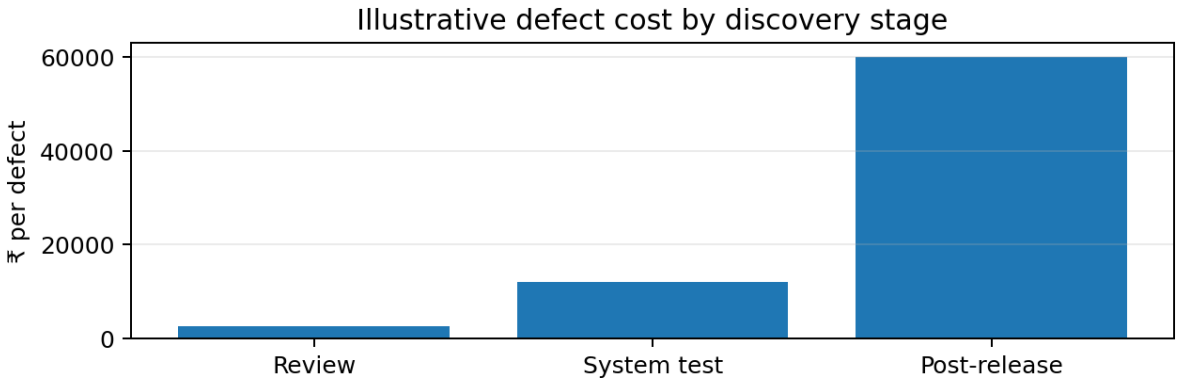


Figure. TASK 6.6 – AI as additional reviewer – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 6.7 – Final review selection rule

This section addresses task 6.7 – final review selection rule for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use formal inspection for high-impact fraud/payout/security logic and lighter reviews elsewhere.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Review	Expected detection	Effort	Use
Desk-check	Low/medium	Low	Routine
Technical	Medium/high	Medium	Complex work
Inspection	High	High	Critical logic

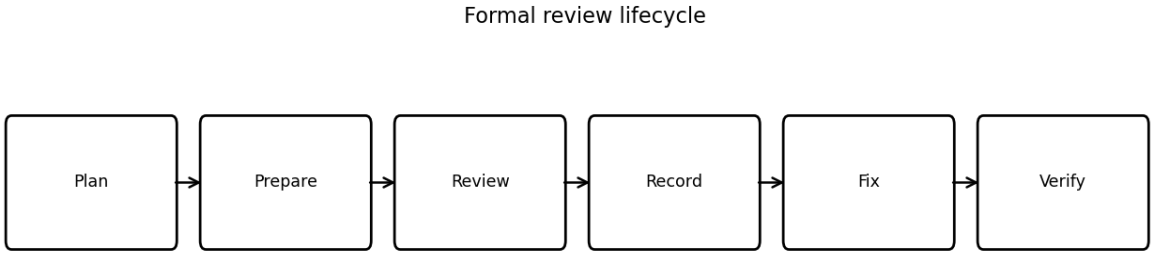


Figure. TASK 6.7 – Final review selection rule – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 7.1 – Prediction and planning comparison

This section addresses task 7.1 – prediction and planning comparison for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use risk-and-size-driven planning anchored to mandatory gates for high-risk artefacts.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Plan	Strength	Weakness	Outcome
Fixed calendar	Simple	Not risk-driven	Variable
Risk/size	Targets value	Needs triage	Best
Gate	Clear sequence	Slip-sensitive	Strong if stable

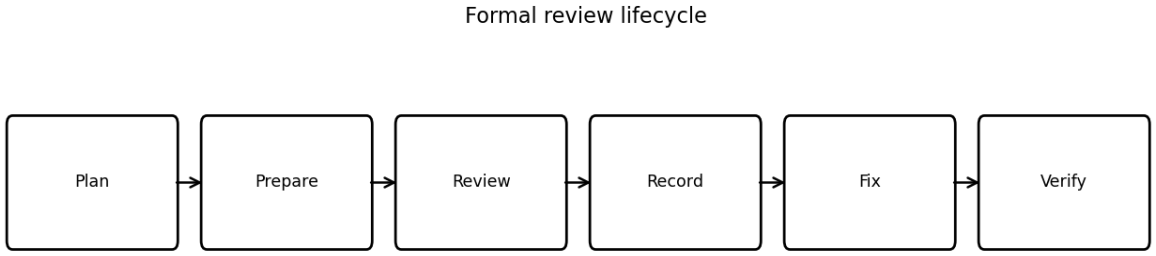


Figure. TASK 7.1 – Prediction and planning comparison – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 7.2 – Project effort and 10% review limit

This section addresses task 7.2 – project effort and 10% review limit for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use risk-and-size-driven planning anchored to mandatory gates for high-risk artefacts.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Calculation	Value
Project effort	5,400 h
10% limit	540 h
22 KLOC/150 lph	146.7 h
Remaining review budget	393.3 h

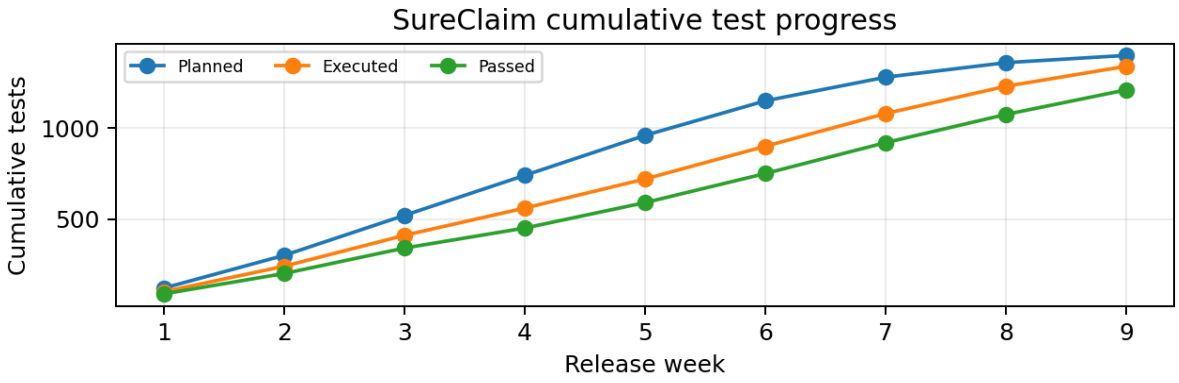


Figure. TASK 7.2 – Project effort and 10% review limit – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 7.3 – Fixed calendar plan

This section addresses task 7.3 – fixed calendar plan for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use risk-and-size-driven planning anchored to mandatory gates for high-risk artefacts.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Gate	Entry	Exit
Requirements	Stable draft	Issues logged
Design	Approved requirements	Decisions recorded
Code	Buildable	Critical issues addressed
Release	Locked candidate	Evidence complete

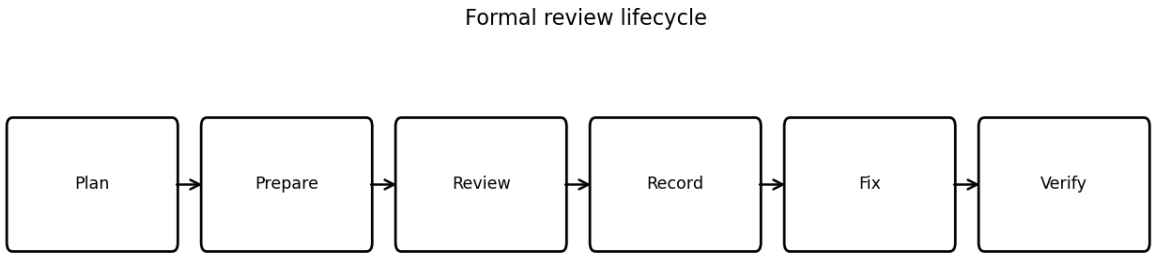


Figure. TASK 7.3 – Fixed calendar plan – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 7.4 – Risk and size-driven plan

This section addresses task 7.4 – risk and size-driven plan for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use risk-and-size-driven planning anchored to mandatory gates for high-risk artefacts.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Plan	Strength	Weakness	Outcome
Fixed calendar	Simple	Not risk-driven	Variable
Risk/size	Targets value	Needs triage	Best
Gate	Clear sequence	Slip-sensitive	Strong if stable

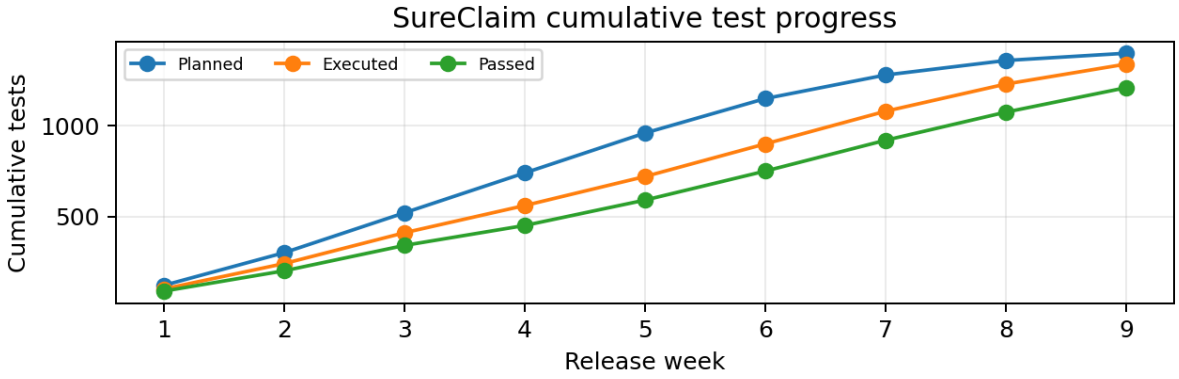


Figure. TASK 7.4 – Risk and size-driven plan – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 7.5 – Gate-based plan and entry/exit criteria

This section addresses task 7.5 – gate-based plan and entry/exit criteria for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use risk-and-size-driven planning anchored to mandatory gates for high-risk artefacts.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Calculation	Value
Project effort	5,400 h
10% limit	540 h
22 KLOC/150 lph	146.7 h
Remaining review budget	393.3 h

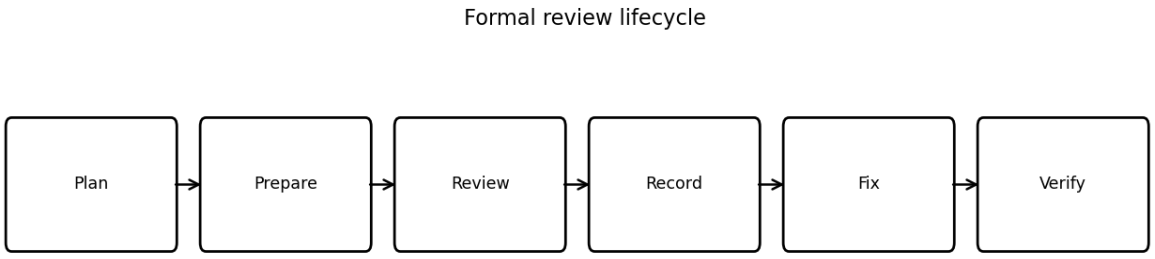


Figure. TASK 7.5 – Gate-based plan and entry/exit criteria – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 7.6 – LLM checklist critique

This section addresses task 7.6 – llm checklist critique for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use risk-and-size-driven planning anchored to mandatory gates for high-risk artefacts.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Gate	Entry	Exit
Requirements	Stable draft	Issues logged
Design	Approved requirements	Decisions recorded
Code	Buildable	Critical issues addressed
Release	Locked candidate	Evidence complete

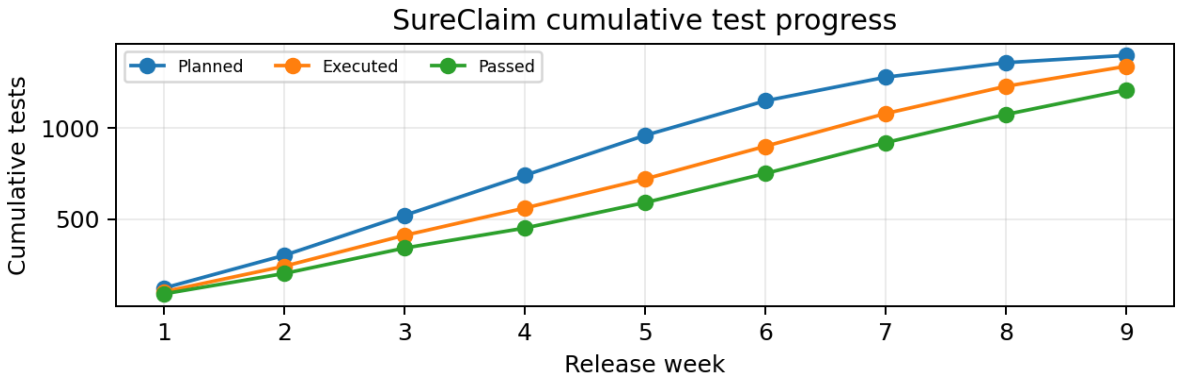


Figure. TASK 7.6 – LLM checklist critique – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 7.7 – Final review plan

This section addresses task 7.7 – final review plan for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use risk-and-size-driven planning anchored to mandatory gates for high-risk artefacts.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Plan	Strength	Weakness	Outcome
Fixed calendar	Simple	Not risk-driven	Variable
Risk/size	Targets value	Needs triage	Best
Gate	Clear sequence	Slip-sensitive	Strong if stable

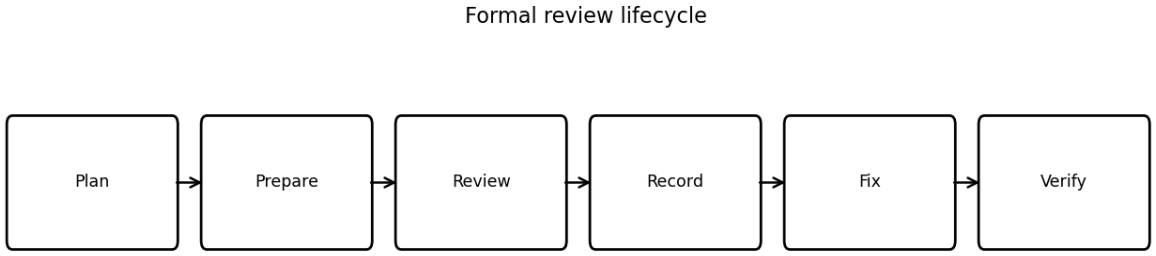


Figure. TASK 7.7 – Final review plan – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 8.1 – Prediction and reporting comparison

This section addresses task 8.1 – prediction and reporting comparison for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use tool-based tracked actions with formal milestone exports.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Format	Traceability	Follow-up	Effort
E-mail	Medium	Weak	Low
Formal	High	Medium/high	Medium
Tool-based	Very high	Very high	Medium

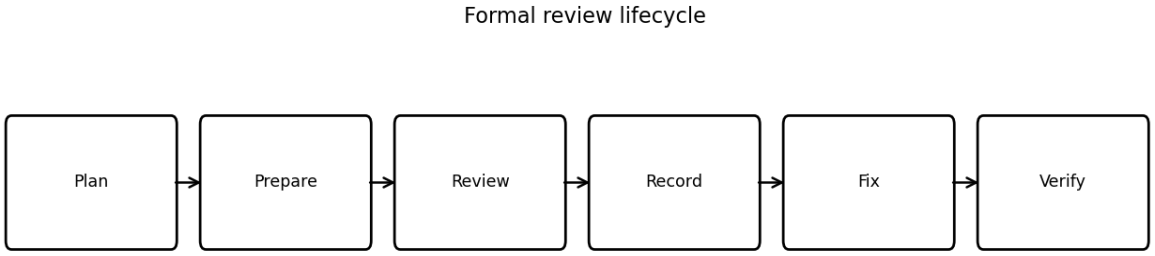


Figure. TASK 8.1 – Prediction and reporting comparison – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 8.2 – Ten-comment review record

This section addresses task 8.2 – ten-comment review record for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use tool-based tracked actions with formal milestone exports.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

ID	Type	Severity	Action
R01	Boundary	Major	Add test
R02	Logic	Critical	Fix
R03	Security	Major	Verify
R04	Requirement	Major	Clarify
R05	Error	Minor	Handle timeout
R06	Logging	Minor	Add reason
R07	Config	Major	Pin version
R08	Testability	Minor	Add deterministic input
R09	Maintainability	Minor	Refactor
R10	Docs	Minor	Update

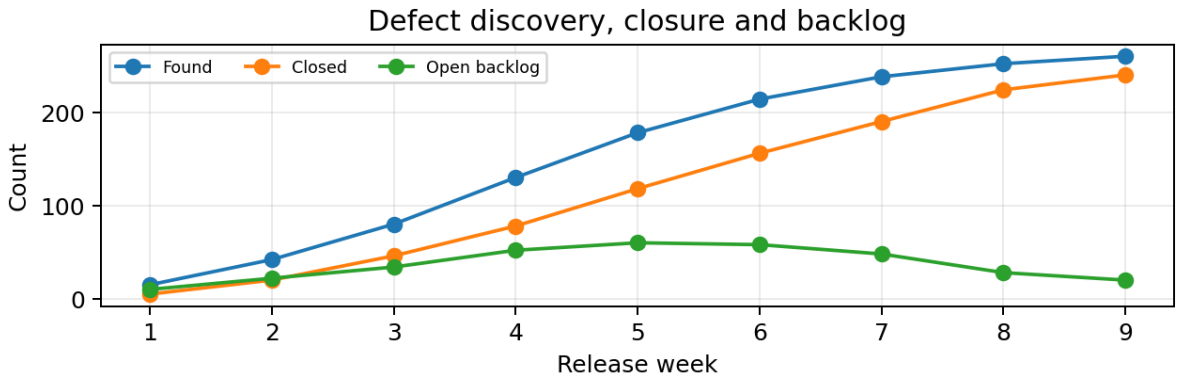


Figure. TASK 8.2 – Ten-comment review record – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 8.3 – Review metrics

This section addresses task 8.3 – review metrics for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use tool-based tracked actions with formal milestone exports.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Metric	Formula	Illustrative result
Defects/hour	10/2	5
Defects/KLOC	10/0.1	100
Major %	4/10	40%

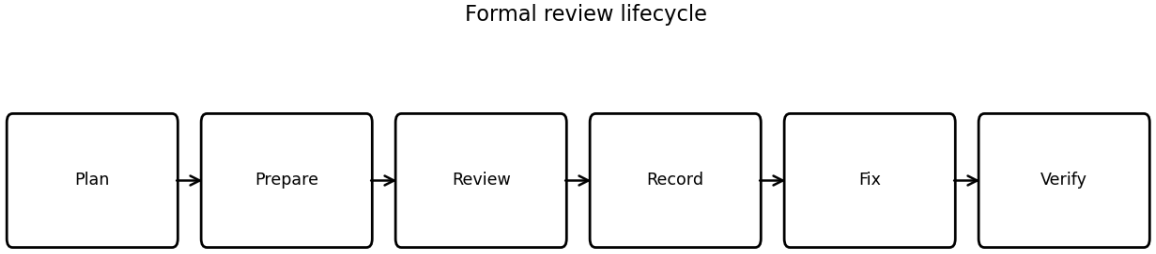


Figure. TASK 8.3 – Review metrics – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 8.4 – Two-week follow-up traceability

This section addresses task 8.4 – two-week follow-up traceability for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use tool-based tracked actions with formal milestone exports.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Format	Traceability	Follow-up	Effort
E-mail	Medium	Weak	Low
Formal	High	Medium/high	Medium
Tool-based	Very high	Very high	Medium

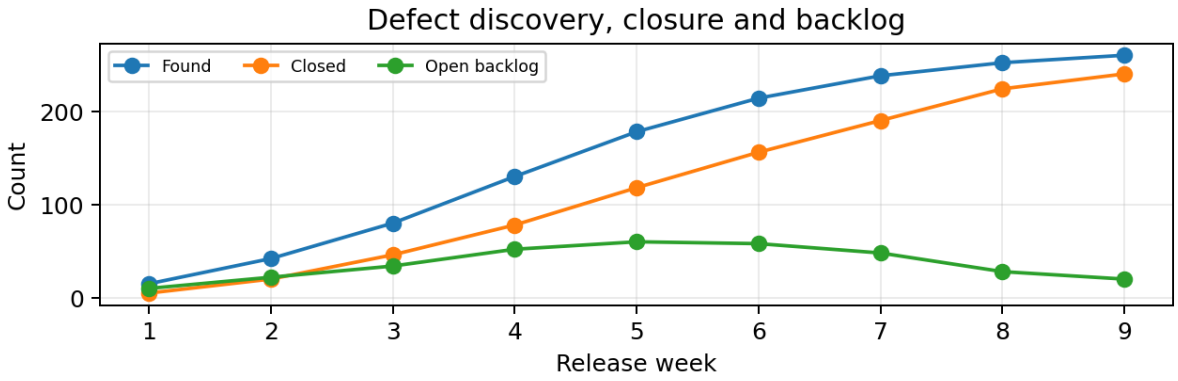


Figure. TASK 8.4 – Two-week follow-up traceability – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 8.5 – Process improvement from review data

This section addresses task 8.5 – process improvement from review data for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use tool-based tracked actions with formal milestone exports.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

ID	Type	Severity	Action
R01	Boundary	Major	Add test
R02	Logic	Critical	Fix
R03	Security	Major	Verify
R04	Requirement	Major	Clarify
R05	Error	Minor	Handle timeout
R06	Logging	Minor	Add reason
R07	Config	Major	Pin version
R08	Testability	Minor	Add deterministic input
R09	Maintainability	Minor	Refactor
R10	Docs	Minor	Update

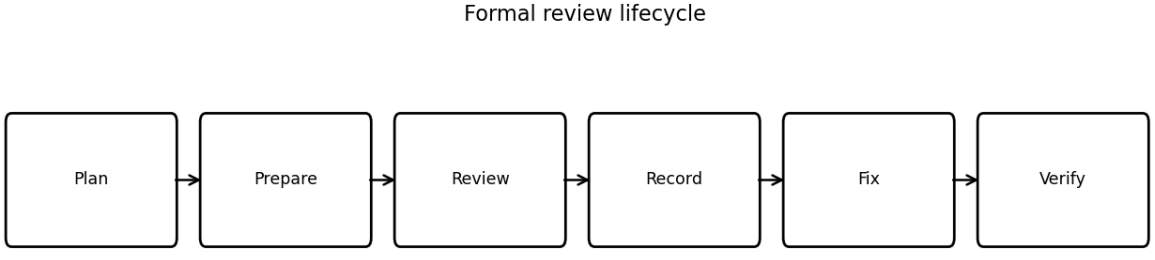


Figure. TASK 8.5 – Process improvement from review data – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 8.6 – Accept-with-changes versus re-review

This section addresses task 8.6 – accept-with-changes versus re-review for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use tool-based tracked actions with formal milestone exports.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Metric	Formula	Illustrative result
Defects/hour	10/2	5
Defects/KLOC	10/0.1	100
Major %	4/10	40%

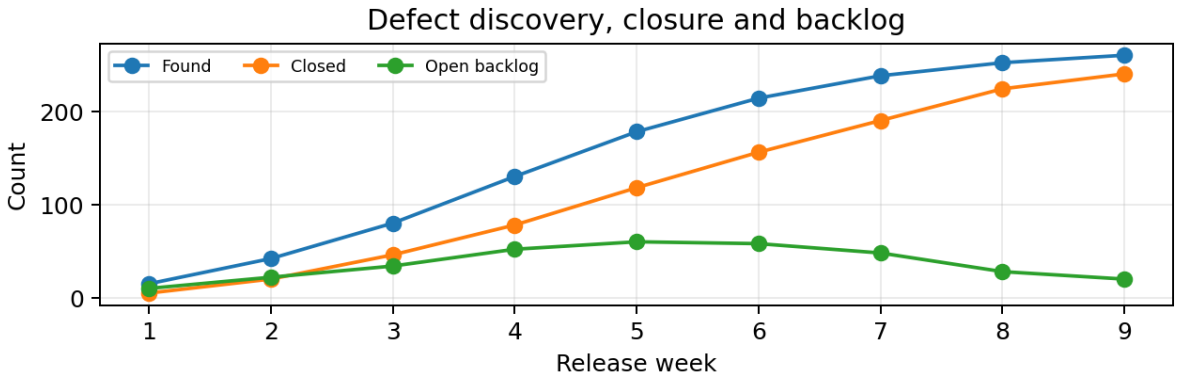


Figure. TASK 8.6 – Accept-with-changes versus re-review – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 8.7 – AI summary reliability and final choice

This section addresses task 8.7 – ai summary reliability and final choice for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Use tool-based tracked actions with formal milestone exports.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Format	Traceability	Follow-up	Effort
E-mail	Medium	Weak	Low
Formal	High	Medium/high	Medium
Tool-based	Very high	Very high	Medium

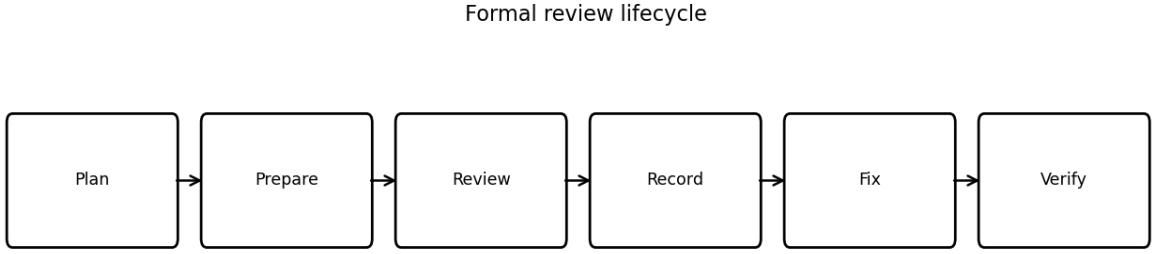


Figure. TASK 8.7 – AI summary reliability and final choice – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 9.1 – Prediction and three-configuration comparison

This section addresses task 9.1 – prediction and three-configuration comparison for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Select the balanced configuration as the baseline and evolve it into quality-as-code as cadence and audit needs increase.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Configuration	Metrics	SCM	Reviews	Stopping
A	Plan/actual	Folder	Informal	Basic
B	Defect+coverage	Git	Risk	Defined
C	Live+audit	Automated	Formal	Risk/cost

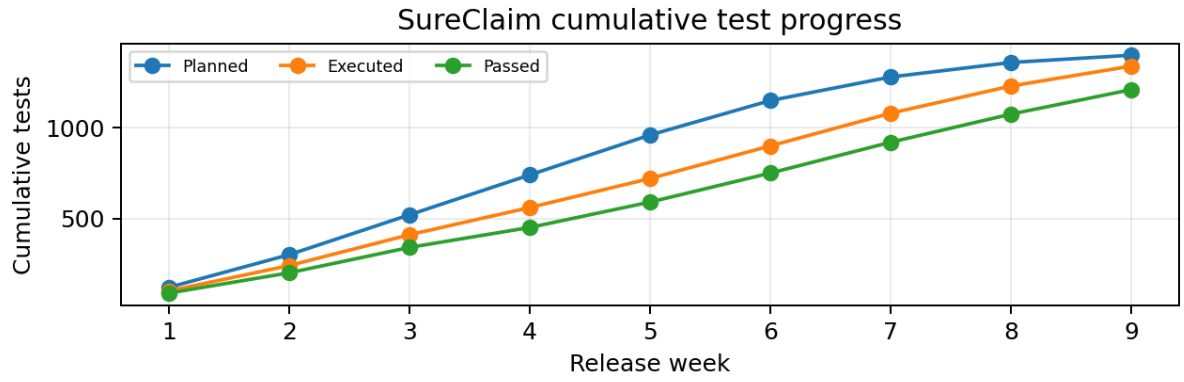


Figure. TASK 9.1 – Prediction and three-configuration comparison – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 9.2 – Metric prototype

This section addresses task 9.2 – metric prototype for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Select the balanced configuration as the baseline and evolve it into quality-as-code as cadence and audit needs increase.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Metric	W5	W9
Execution	75.0%	95.7%
Pass	81.9%	90.3%
Variance	−240	−60
Open	60	20
Density	8.09	11.82

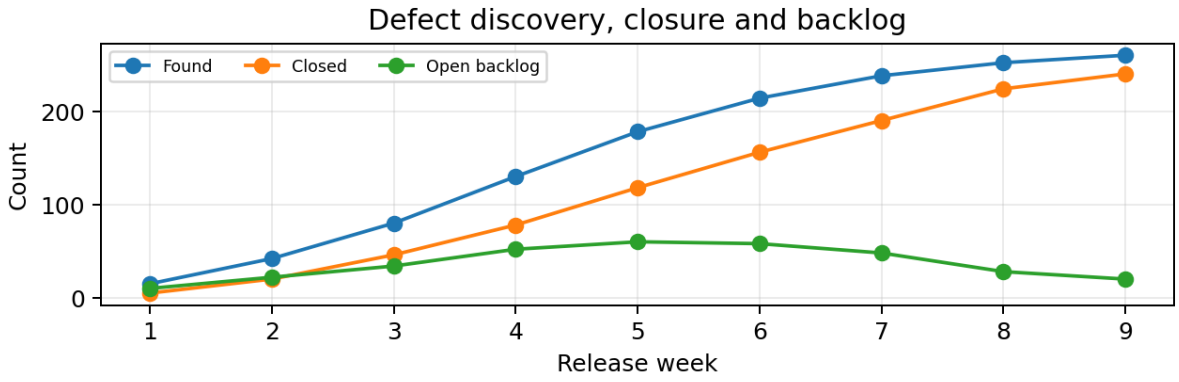


Figure. TASK 9.2 – Metric prototype – supporting visual

Illustrative implementation code is kept left-aligned. In an actual exercise, the source file and generated output should be archived with the test/build identifier.

```
def metric(planned, executed, passed, found, closed, kloc=22):
    return {
        "execution_pct": executed / planned * 100,
```

```
        "pass_pct": passed / executed * 100,
        "variance": executed - planned,
        "open": found - closed,
        "density": found / kloc
    }
```

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 9.3 – RAG and stopping logic

This section addresses task 9.3 – rag and stopping logic for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Select the balanced configuration as the baseline and evolve it into quality-as-code as cadence and audit needs increase.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Dimension	A	B	C
Early warning	Low	High	Very high
Auditability	Low	High	Very high
Cost	Low	Medium	High
Release confidence	Low	High	Very high

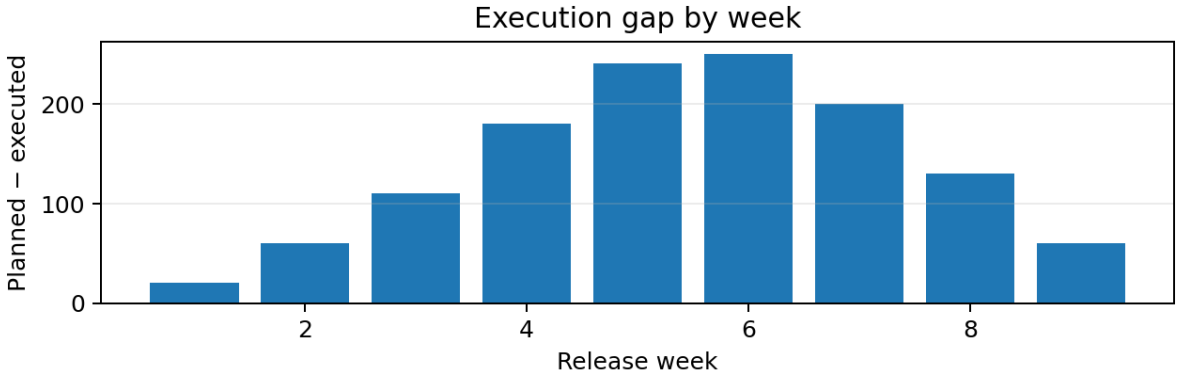


Figure. TASK 9.3 – RAG and stopping logic – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 9.4 – Minimal versus balanced versus comprehensive configuration

This section addresses task 9.4 – minimal versus balanced versus comprehensive configuration for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Select the balanced configuration as the baseline and evolve it into quality-as-code as cadence and audit needs increase.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Configuration	Metrics	SCM	Reviews	Stopping
A	Plan/actual	Folder	Informal	Basic
B	Defect+coverage	Git	Risk	Defined
C	Live+audit	Automated	Formal	Risk/cost

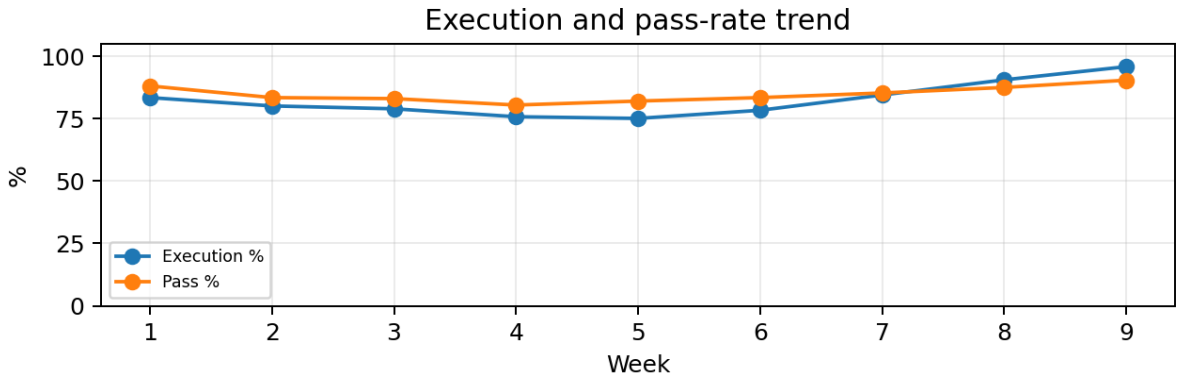


Figure. TASK 9.4 – Minimal versus balanced versus comprehensive configuration – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 9.5 – Design decision table

This section addresses task 9.5 – design decision table for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Select the balanced configuration as the baseline and evolve it into quality-as-code as cadence and audit needs increase.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Metric	W5	W9
Execution	75.0%	95.7%
Pass	81.9%	90.3%
Variance	−240	−60
Open	60	20
Density	8.09	11.82

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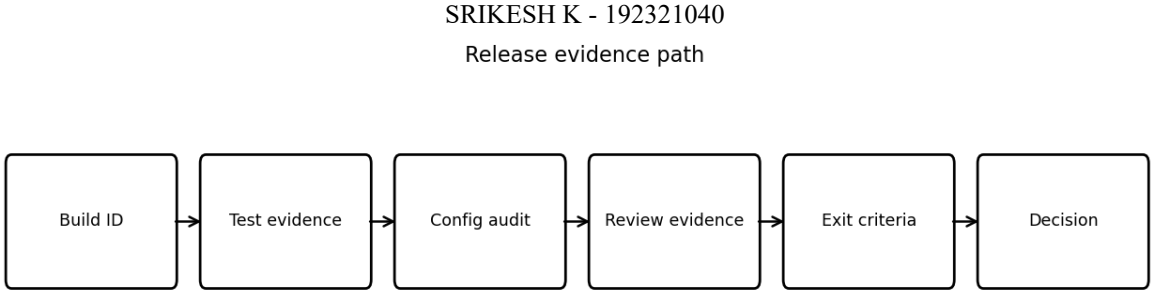


Figure. TASK 9.5 – Design decision table – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 9.6 – Transfer to ATM, hospital pharmacy and mobile game

This section addresses task 9.6 – transfer to atm, hospital pharmacy and mobile game for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Select the balanced configuration as the baseline and evolve it into quality-as-code as cadence and audit needs increase.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Dimension	A	B	C
Early warning	Low	High	Very high
Auditability	Low	High	Very high
Cost	Low	Medium	High
Release confidence	Low	High	Very high

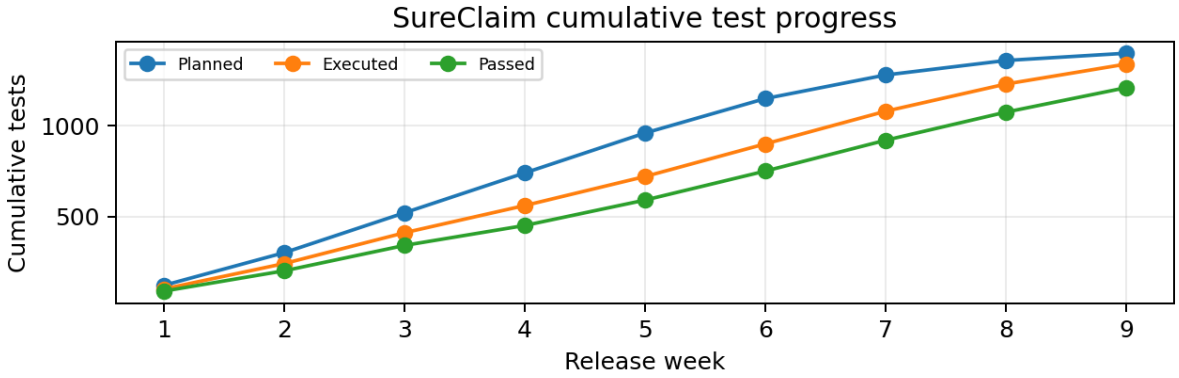


Figure. TASK 9.6 – Transfer to ATM, hospital pharmacy and mobile game – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 9.7 – Cumulative-data limitation and contradictory evidence

This section addresses task 9.7 – cumulative-data limitation and contradictory evidence for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Select the balanced configuration as the baseline and evolve it into quality-as-code as cadence and audit needs increase.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Configuration	Metrics	SCM	Reviews	Stopping
A	Plan/actual	Folder	Informal	Basic
B	Defect+coverage	Git	Risk	Defined
C	Live+audit	Automated	Formal	Risk/cost

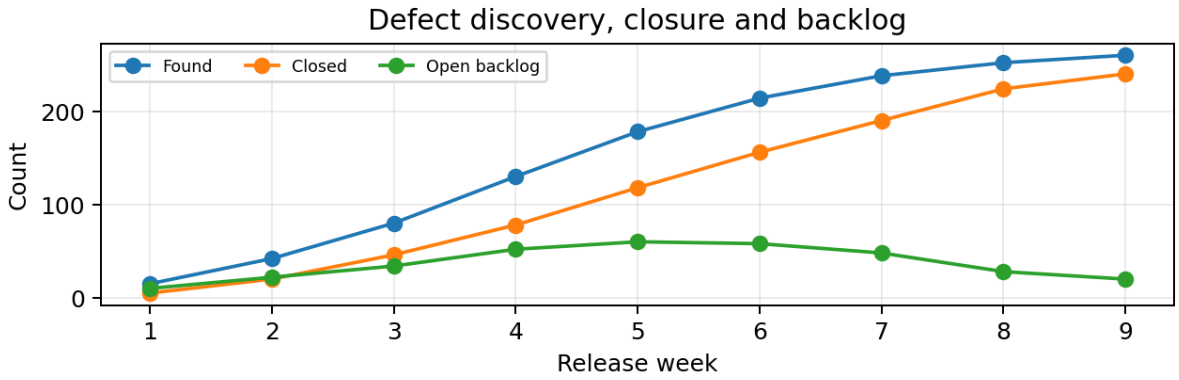


Figure. TASK 9.7 – Cumulative-data limitation and contradictory evidence – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 9.8 – Fourth AI-assisted configuration

This section addresses task 9.8 – fourth ai-assisted configuration for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: Select the balanced configuration as the baseline and evolve it into quality-as-code as cadence and audit needs increase.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Metric	W5	W9
Execution	75.0%	95.7%
Pass	81.9%	90.3%
Variance	−240	−60
Open	60	20
Density	8.09	11.82

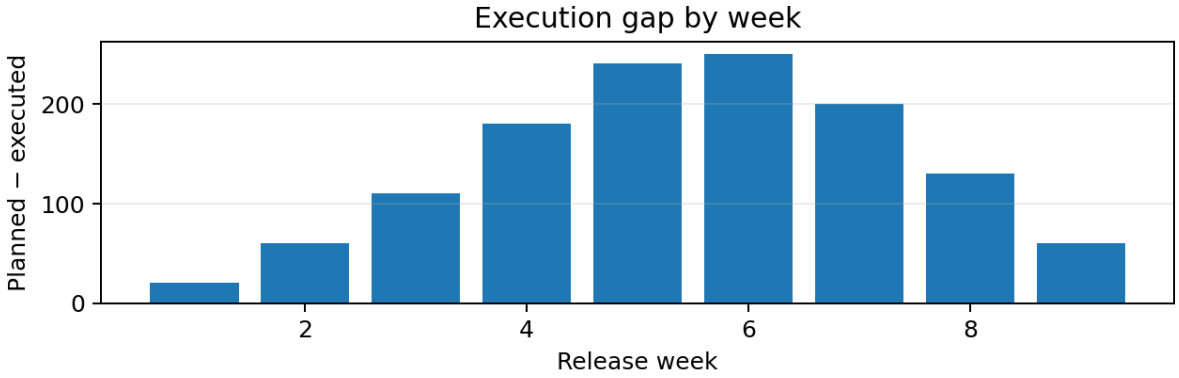


Figure. TASK 9.8 – Fourth AI-assisted configuration – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 10.1 – Changed scenario and prediction

This section addresses task 10.1 – changed scenario and prediction for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: REDESIGN toward quality-as-code, implemented first through pipeline gates and tracked lightweight reviews.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Changed factor	Original	New	Impact
Release	12 weeks	2 weeks	Automation
Team	15	24 incl. vendors	Coordination
Payout	Existing	UPI ≤30 min	Critical path
Fraud	Internal	External AI	Dependency
Overhead	10% review	≤8% total	Lean process

Quality-as-code pipeline

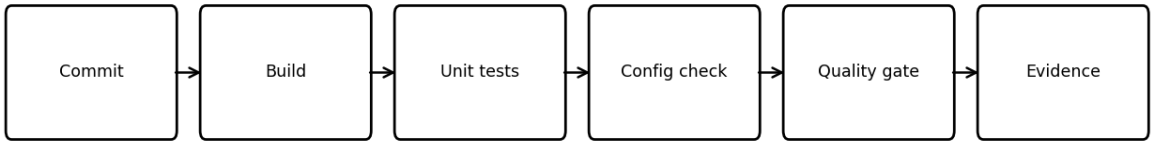


Figure. TASK 10.1 – Changed scenario and prediction – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 10.2 – Strategy A: retain and scale down

This section addresses task 10.2 – strategy a: retain and scale down for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: REDESIGN toward quality-as-code, implemented first through pipeline gates and tracked lightweight reviews.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Strategy	Overhead	Audit	Decision
A	Low/medium	Medium	Reject
B	Medium	High	Immediate
C	Higher setup	Very high	Target state

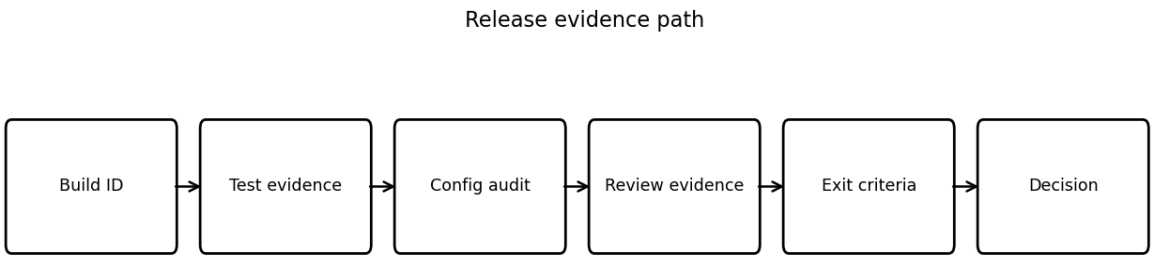


Figure. TASK 10.2 – Strategy A: retain and scale down – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 10.3 – Strategy B: pipeline gates and lightweight reviews

This section addresses task 10.3 – strategy b: pipeline gates and lightweight reviews for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: REDESIGN toward quality-as-code, implemented first through pipeline gates and tracked lightweight reviews.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Calculation	Value
Team	24
Cycle	2 weeks
Hours/person/week	30
Cycle effort	1,440 h
8% ceiling	115.2 h

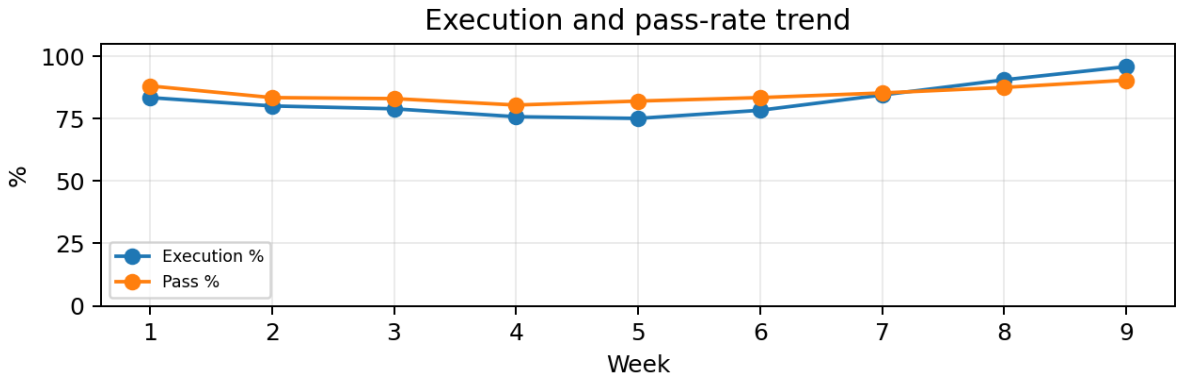


Figure. TASK 10.3 – Strategy B: pipeline gates and lightweight reviews – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 10.4 – Strategy C: quality-as-code

This section addresses task 10.4 – strategy c: quality-as-code for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: REDESIGN toward quality-as-code, implemented first through pipeline gates and tracked lightweight reviews.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Evidence	Frequency	Owner
Build/config	Every deployment	Build engineer
Critical tests	Every pipeline run	Test lead
Defects	Daily	QA analyst
Reviews	Daily	Review owners
Release decision	Fortnightly	Release manager

Quality-as-code pipeline

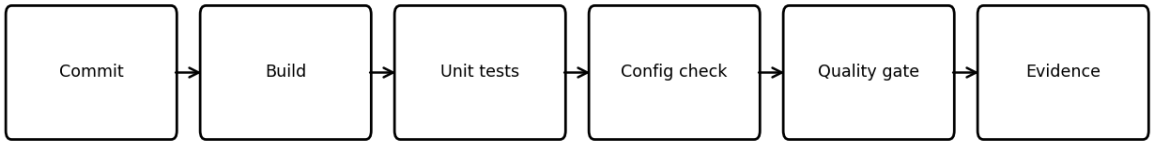


Figure. TASK 10.4 – Strategy C: quality-as-code – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 10.5 – Two-week overhead calculation

This section addresses task 10.5 – two-week overhead calculation for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: REDESIGN toward quality-as-code, implemented first through pipeline gates and tracked lightweight reviews.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Changed factor	Original	New	Impact
Release	12 weeks	2 weeks	Automation
Team	15	24 incl. vendors	Coordination
Payout	Existing	UPI ≤30 min	Critical path
Fraud	Internal	External AI	Dependency
Overhead	10% review	≤8% total	Lean process

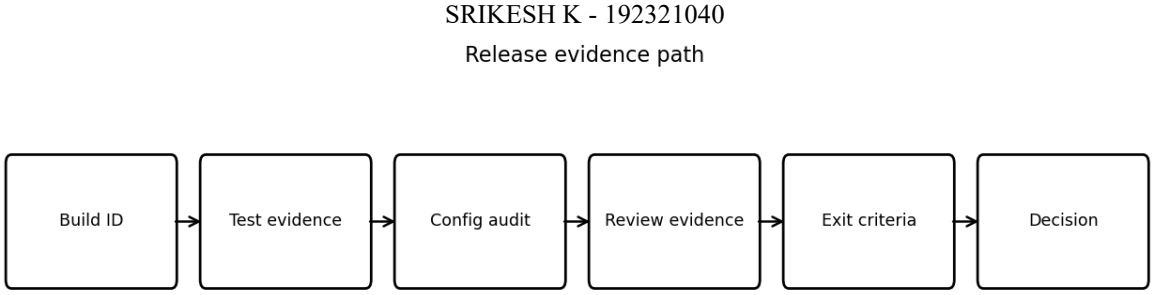


Figure. TASK 10.5 – Two-week overhead calculation – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 10.6 – Vendor and external AI risk

This section addresses task 10.6 – vendor and external ai risk for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: REDESIGN toward quality-as-code, implemented first through pipeline gates and tracked lightweight reviews.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Strategy	Overhead	Audit	Decision
A	Low/medium	Medium	Reject
B	Medium	High	Immediate
C	Higher setup	Very high	Target state

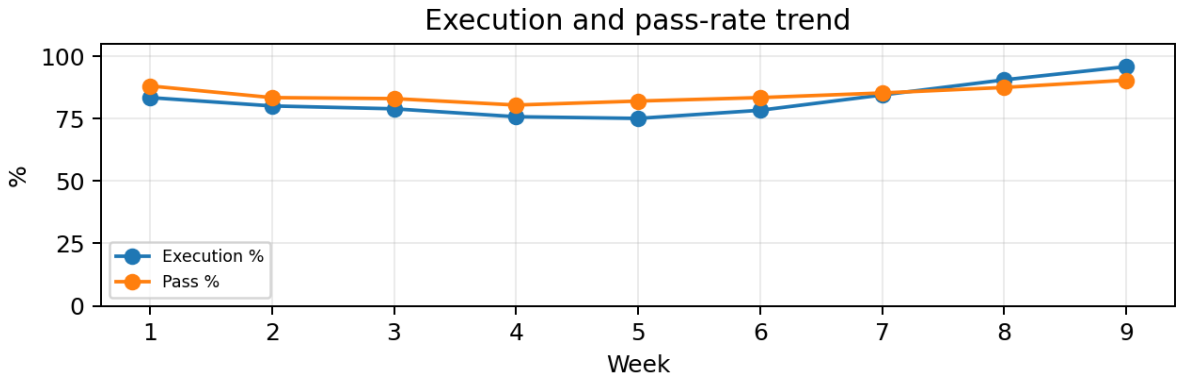


Figure. TASK 10.6 – Vendor and external AI risk – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 10.7 – Revised daily/release evidence set

This section addresses task 10.7 – revised daily/release evidence set for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: REDESIGN toward quality-as-code, implemented first through pipeline gates and tracked lightweight reviews.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Calculation	Value
Team	24
Cycle	2 weeks
Hours/person/week	30
Cycle effort	1,440 h
8% ceiling	115.2 h

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Quality-as-code pipeline

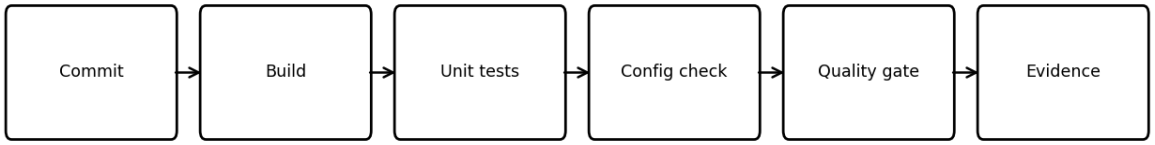


Figure. TASK 10.7 – Revised daily/release evidence set – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

TASK 10.8 – Final RETAIN/MODIFY/REDESIGN decision

This section addresses task 10.8 – final retain/modify/redesign decision for the SureClaim v2.0 release. The supplied baseline contains 22 KLOC of new or changed code, 96 requirements or user stories, 1,400 planned test cases, a 12-week release and a 15-person team. The analysis distinguishes supplied evidence from explicit planning assumptions. This distinction is important because a professional quality report must never present a simulated result as if it were a measured production observation.

Prediction for this subtask: REDESIGN toward quality-as-code, implemented first through pipeline gates and tracked lightweight reviews.. The comparison is performed against the three techniques named in the assignment, and the chosen technique is judged against evidence, effort, traceability and its failure mode. Every candidate technique is considered on effectiveness, cost, traceability and failure mode. The preferred choice is the one that provides a defensible management decision with the least unnecessary process burden. Where an exercise requires execution that is not contained in the supplied data, the report provides a reproducible prototype or controlled illustrative result and clearly identifies the assumption.

Evidence	Frequency	Owner
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Defects	Daily	QA analyst
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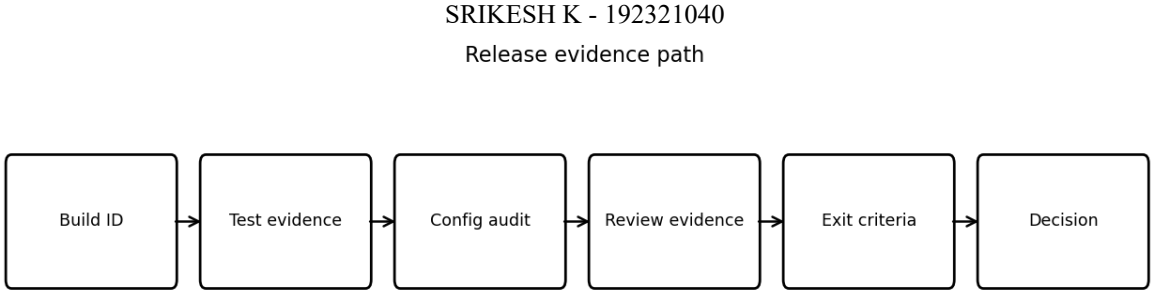


Figure. TASK 10.8 – Final RETAIN/MODIFY/REDESIGN decision – supporting visual

Control conclusion: the selected practice should be linked to an owner, a measurable threshold, a source record and an explicit response. If the control is omitted, the team may still perform testing or reviews, but management loses the evidence needed to distinguish genuine quality improvement from changes in workload, configuration, process or reporting.

APPENDIX A – WEEKLY TRACKING DATA

The cumulative dataset should be interpreted carefully. A cumulative execution count tells management how much work has been completed by a point in time, but it does not by itself explain why the team is behind or whether the remaining tests are concentrated in high-risk areas. For that reason, the cumulative charts in this report are supplemented by variance, pass-rate, defect-flow and coverage indicators.

The week-to-week changes are also useful for understanding recovery. Execution increases rapidly after the early schedule gap, while defect closure accelerates in the later weeks. However, the presence of three critical defects at week 9 means that schedule recovery does not automatically mean release readiness.

A real project dashboard should preserve the historical snapshots instead of overwriting the current values. Historical snapshots allow the test manager to determine whether a red condition was improving, stable or deteriorating and whether the corrective action produced the expected effect.

The dataset should therefore be treated as a baseline for decision-making, not as a substitute for detailed test records. Individual test execution records, defect IDs, severity, configuration identity and requirement links remain necessary for audit and reproduction.

Source data used throughout the report

Week	Planned	Executed	Passed	Found	Closed	Open
1	120	100	88	15	5	10
2	300	240	200	42	20	22
3	520	410	340	80	46	34
4	740	560	450	130	78	52
5	960	720	590	178	118	60
6	1150	900	750	214	156	58
7	1280	1080	920	238	190	48
8	1360	1230	1075	252	224	28
9	1400	1340	1210	260	240	20

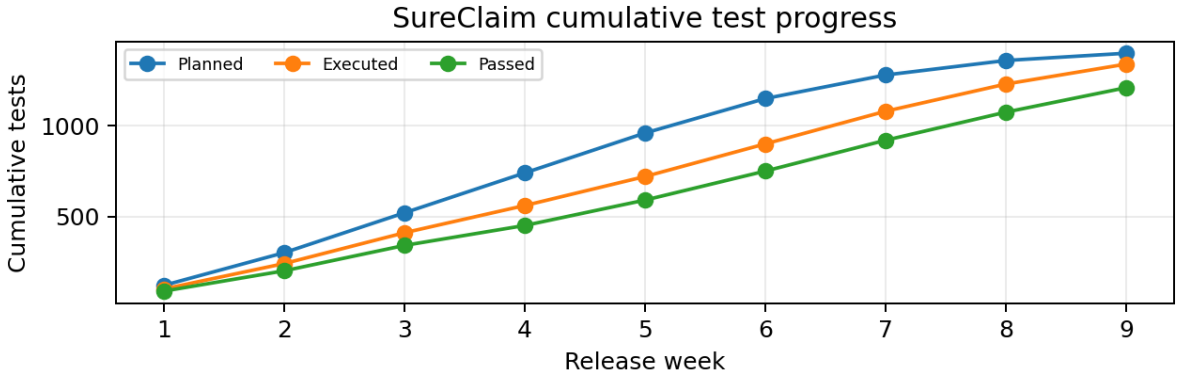


Figure. Cumulative planned, executed and passed tests

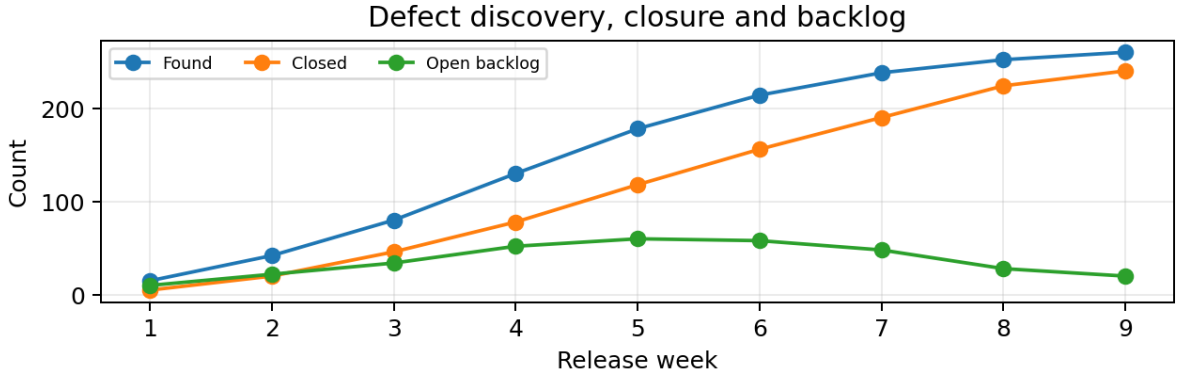


Figure. Cumulative defects found, closed and open backlog

The values are cumulative. Weekly new defects are obtained by differencing the cumulative found series: 15, 27, 38, 50, 48, 36, 24, 14 and 8. Weekly closures are 5, 15, 26, 32, 40, 38, 34, 34 and 16. The data does not independently show retest volume or severity by week, so those aspects must be obtained from the test and defect tools in a real release.

APPENDIX B – METRIC AND RAG CALCULATION SHEET

Metric definitions should be frozen before the reporting cycle begins. If the denominator or inclusion rule changes after a poor result, the apparent improvement may be artificial. For example, removing blocked tests from the planned count can make execution appear healthier without completing any additional testing.

RAG thresholds should also be interpreted in context. A green schedule indicator cannot override a mandatory quality gate. Similarly, a high pass rate can coexist with a serious risk if the failed tests are concentrated in payment, authentication, fraud or other high-impact functionality.

Each red or amber result should have an associated action. The action should state the cause being investigated, the owner, the expected completion date and the evidence that will show whether the response worked. This makes monitoring a control mechanism rather than a passive reporting exercise.

Reusable formulas

Metric	Formula	W5	W9
Execution %	Executed/Planned	75.00%	95.71%
Pass %	Passed/Executed	81.94%	90.30%
Variance	Executed−Planned	−240	−60
Find rate	Found/Executed	24.72%	19.40%
Closure	Closed/Found	66.29%	92.31%
Open	Found−Closed	60	20
Density	Found/22	8.09	11.82/KLOC

```
def rag(planned, executed, critical_open, config_ok):
    gap = (planned - executed) / planned
    if critical_open > 0 or not config_ok:
        return "RED"
    if gap > 0.15:
        return "RED"
    if gap > 0.05:
        return "AMBER"
```

```
    return "GREEN"
```

The same formulas can be implemented in a spreadsheet, Python or a test-management tool. The definitions should be baselined before reporting starts so that a metric does not silently change meaning between weeks.

APPENDIX C – REVIEW, CONFIGURATION AND RELEASE CHECKLIST

The checklist is intended to be used as an operational control rather than as a decorative appendix. Before a build is promoted, the responsible owner should mark each control as complete, incomplete, or not applicable and attach the corresponding evidence. A control marked incomplete must have an owner and target date; otherwise the release decision should treat the missing evidence as residual risk.

Measurement evidence should be generated from the same source used for the management dashboard. This avoids a situation in which the report shows one value while the test tool contains another. The release manager should verify the timestamp, build identifier and reporting period before accepting the dashboard as evidence.

Configuration evidence is particularly important because a successful test run has little value if the tested configuration cannot be proven to be the candidate released to consumers. The baseline should therefore identify source revision, configuration files, tariff or rule tables, test-data version and deployment package. The comparison should be reproducible by another team member.

Review evidence should show not only that a review occurred but also what was reviewed, who participated, what defects or risks were identified, how each action was resolved and whether the resolution was independently checked. This converts a meeting into an auditable quality activity.

Release evidence should be assembled before the final approval meeting. The evidence pack should contain the latest test summary, open-defect list, coverage snapshot, configuration manifest, review status, risk acceptance and any approved exceptions. A green status without this supporting evidence should not be treated as a release recommendation.

These controls also support future audits. When a production incident occurs, the organisation can reconstruct the candidate, determine which tests were executed, identify the reviews that were performed and understand why the release was considered acceptable at that time. This is more valuable than simply storing a final pass percentage.

Practical evidence checklist

Control	Question	Evidence
Measurement	Are definitions frozen?	Metric specification
Status	Does RAG use agreed thresholds?	Weekly snapshot
Monitoring	Is a deteriorating trend assigned?	Action record
Stopping	Are critical gates explicit?	Exit criteria
Configuration	Is candidate identity known?	Manifest/hash
Review	Are high-risk artefacts reviewed?	Review report
Follow-up	Are actions verified?	Tool links
Release	Is residual risk accepted?	Signed decision
AI	Are outputs checked?	Human review record

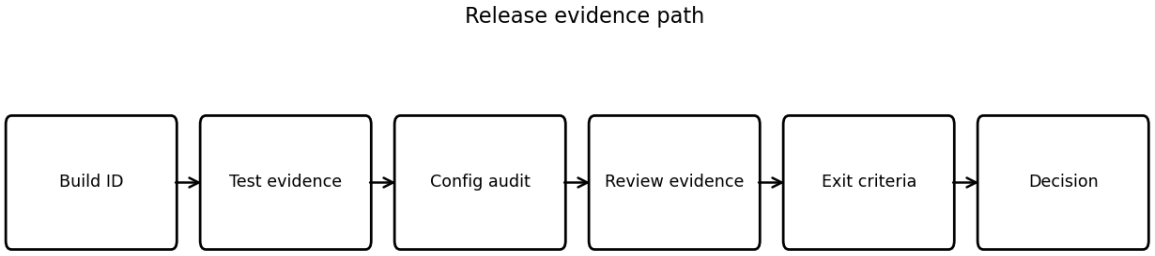


Figure. Release evidence path

A green dashboard does not override a failed mandatory gate. The release decision must verify candidate identity, configuration audit, critical defects, required coverage, review completion and residual-risk acceptance. The same checklist can be tailored for lower-risk products by reducing review depth while retaining identity and accountability.

The release evidence path should be treated as a sequence of independent checks. A build can have a strong pass rate and still fail the configuration gate; a correctly configured build can still fail the critical-defect gate; and a technically acceptable candidate can still require risk acceptance if a lower-priority test has been deferred. Separating these decisions prevents one attractive metric from hiding a serious weakness.

For audit purposes, every evidence item should have four properties: identity, timestamp, owner and source. Identity establishes which requirement, test, build or defect is being discussed. The timestamp establishes the reporting period. The owner identifies

who was accountable for the activity. The source allows an auditor or another engineer to reproduce the evidence rather than relying on a copied number.

The checklist also provides a practical handover mechanism. If the test manager is absent, another authorised person can inspect the same evidence and understand the current release position. This reduces dependence on personal memory and prevents important quality information from remaining in private spreadsheets, email conversations or informal meetings.

A mature process should periodically review the checklist itself. Controls that are consistently completed without influencing decisions may be candidates for automation or simplification, while controls associated with escaped defects or configuration failures should become stronger. The objective is not to increase paperwork; it is to ensure that the smallest useful set of evidence provides a reliable basis for release decisions.

APPENDIX D – FINAL DESIGN DECISION AND SIGN-OFF

The final design brings the separate activities of measurement, reporting, monitoring, configuration management and review into a single quality-control loop. Measurement provides the facts; reporting communicates those facts; monitoring detects deterioration; control actions address the cause; configuration management establishes exactly what was tested; and reviews provide an additional human check over high-impact work products.

The recommended process is deliberately balanced. A process that collects every possible metric, requires every document for every change and reviews every line of work would consume excessive effort and encourage teams to bypass the process. Conversely, a very lightweight process would make it difficult to detect a wrong build, an untested critical requirement or a growing defect backlog. The selected controls therefore become stronger as risk increases.

For SureClaim, the highest-value controls are the zero-open-critical-defect release gate, requirement and coverage evidence for high-risk functionality, controlled candidate identification, tracked review actions and a reproducible release evidence package. These controls directly address the weaknesses revealed by the supplied project data: schedule delay, incomplete coverage, remaining critical defects and the need to justify release readiness.

The process should be reviewed after the release. The team should compare predicted and observed metrics, identify which warnings were useful, determine which reports were ignored, and calculate the effort spent on evidence. The resulting lessons should be fed into the next release rather than remaining as a one-time retrospective.

Final approval should be based on evidence rather than optimism. If a mandatory gate fails, the appropriate response is to hold the release, record the reason, identify the responsible owner and define the conditions for re-evaluation. If an exception is accepted, the decision-maker and residual risk should be explicitly recorded.

The same design can scale to a faster release cadence. As SureClaim moves toward fortnightly delivery, manual evidence collection should progressively be replaced by pipeline-generated test results, signed artifacts, configuration manifests, automated quality gates and dashboard snapshots. Human review remains essential for risk acceptance and business-impact decisions.

Recommended SureClaim practice

Decision	Selected control	Why	If omitted
Metrics	Balanced ≤8	Early warning + risk	Ambiguous progress
Status	RAG + dashboard	Audience fit	Long mail ignored
Control	Remove causes + capacity	Sustainable flow	Late effort masks bottleneck
Stopping	Risk/defect/cost	No false completion	Premature release
SCM	Git + baseline + audit	Correct candidate	Wrong config risk
Reviews	Risk-driven + inspection	Value within limit	High-impact escape
Reporting	Tracked actions	Closure visible	Oral comments lost
Adaptation	Quality-as-code	Fortnightly evidence	Manual overhead

Final recommendation: adopt the balanced configuration immediately and evolve it into quality-as-code as the release cadence accelerates. The release manager owns the release decision; the build/configuration engineer owns candidate identity; testers own execution evidence; developers own fixes; reviewers own review decisions; and the QA analyst owns metric integrity. AI may accelerate analysis, but it must not autonomously approve a release or alter production configuration.

Student: SRIKESH K | Register No.: 192321040 | Department: Information Technology | Faculty decision: GO / HOLD | Signature: _____ | Date: _____

The recommended operating rhythm is therefore: collect evidence continuously, review the dashboard at least weekly, investigate red or deteriorating trends immediately, perform risk-based reviews before major milestones, verify the candidate configuration before release, and preserve the final evidence package. This rhythm gives the Chief Engineer a concise decision view while still giving testers and developers enough detail to investigate individual failures.

The main trade-off is between completeness and speed. The process does not require every possible test, review or report to be performed at maximum depth. Instead, it protects the highest-consequence areas and reduces documentation effort for routine, low-risk work. This is the most sustainable approach because the process remains usable during periods of schedule pressure.

One important failure condition is a late regulatory or business-rule change that affects billing, claims, payout or fraud logic after the candidate has already passed the normal gates. In that situation the process should temporarily increase review and regression scope for the affected requirements, regenerate the configuration baseline and repeat the relevant release gates before approval.

The final sign-off should consequently be interpreted as an evidence-based quality judgement, not a statement that the product contains no defects. The software remains subject to operational monitoring after release. Production incidents, escaped defects and unexpected user behaviour should be fed back into the measurement and review process so that future releases become more predictable.

Sign-off is meaningful only after the evidence pack has been checked against the approved build and the latest test and review records. The signature confirms that the decision-maker has considered the known residual risks; it does not imply that software is defect-free. Any accepted exception should be recorded with its rationale, owner and expiry or review condition.